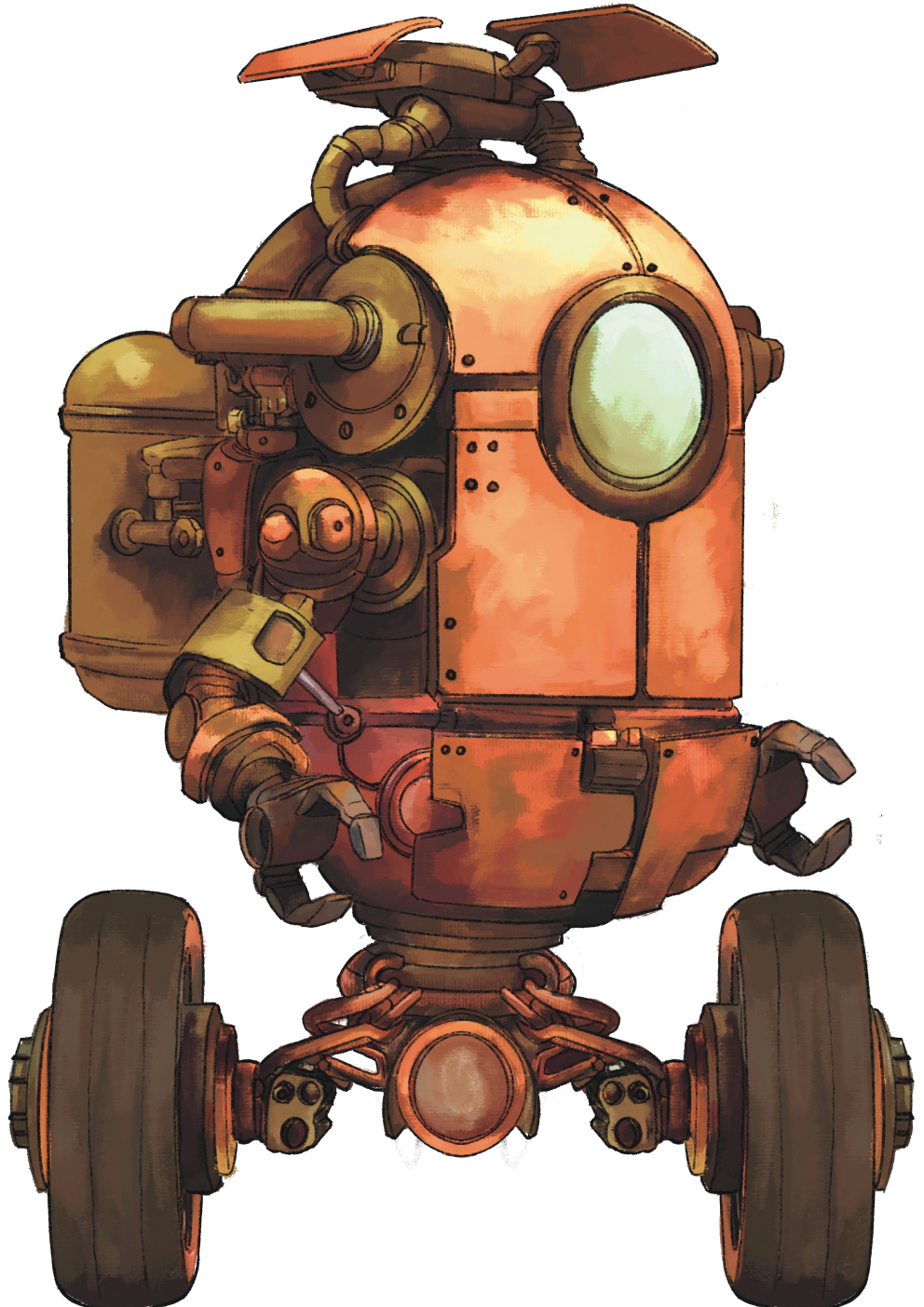
CS 3630, Fall 2026

Lecture 19 (Section 5.4):

Computer Vision and Convolutional Neural Networks



Motivation

- Robotics:
  - Perception, thinking, acting
- Deep learning has revolutionized perception
- Getting increasingly important in thinking/acting
- This lecture:
  - Basic Image Filtering
  - High-level intro to CNNs
  - Applications in robotics
- More resources:
  - Computer vision courses at Georgia Tech
  - Deep Learning courses at Georgia Tech
  - Great online resource:



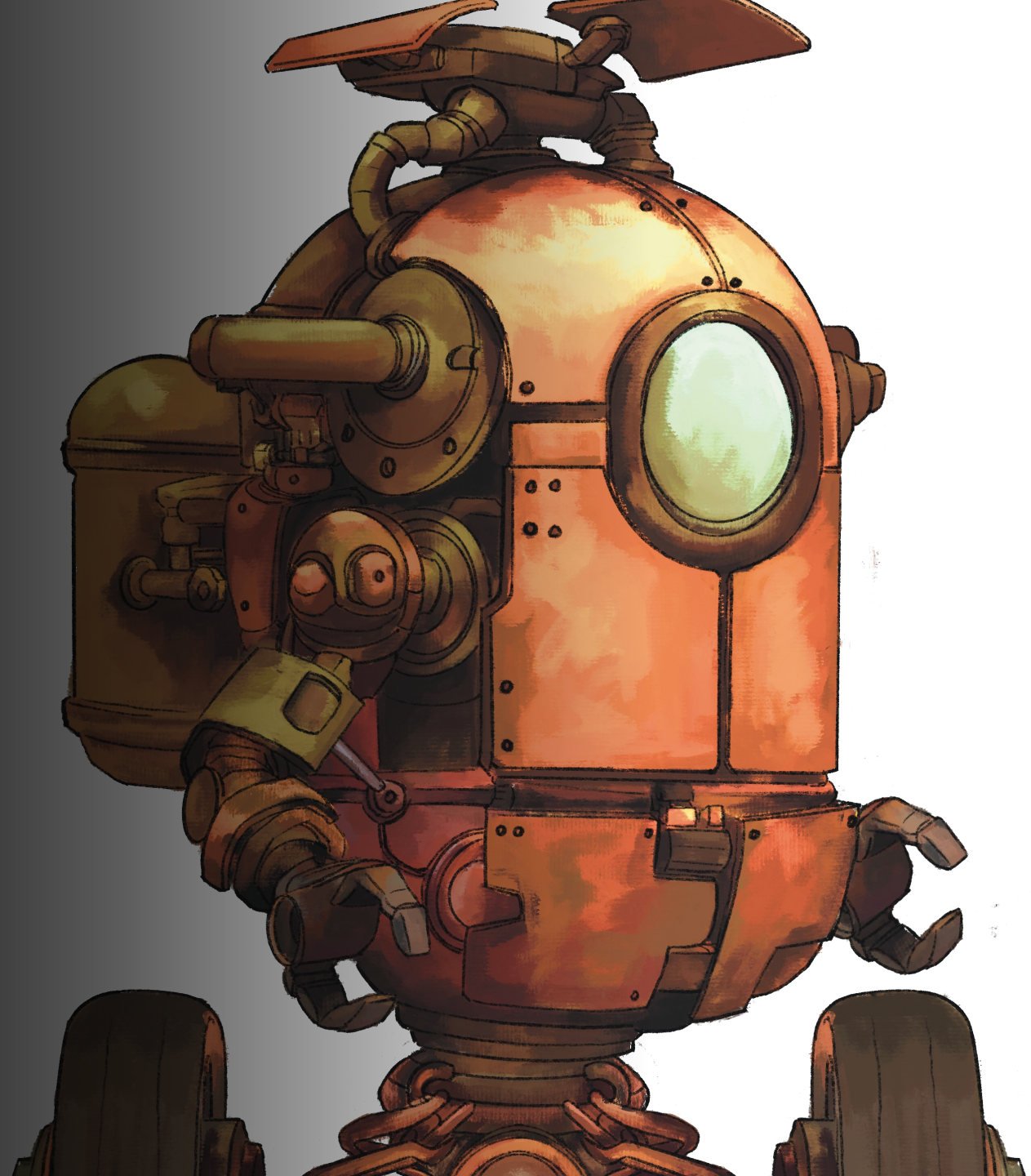
Image by Voyage



https://d2l.ai

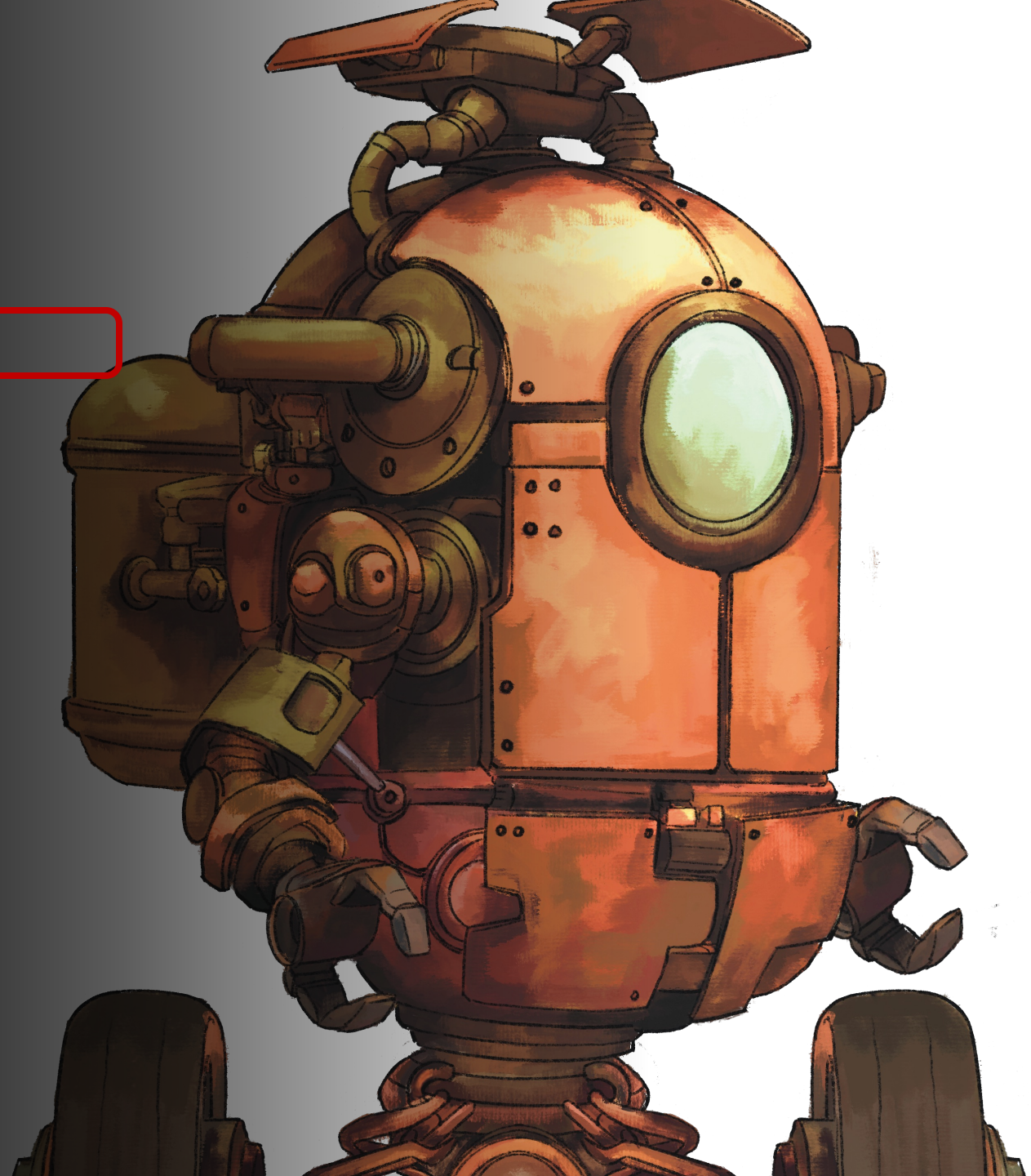
Outline

5. Convolutional Neural Nets
6. Visualizing Learned Filters
7. Pooling and FC Layers
8. Applications in Vision
9. Object Classification
10. Applications in Robotics

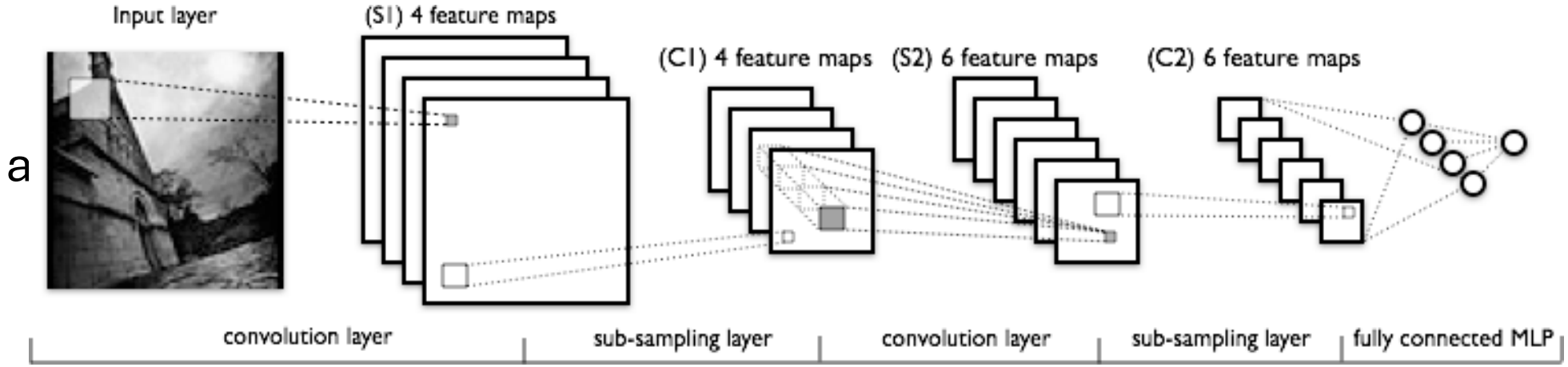


Outline

5. Convolutional Neural Nets
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7. Pooling and FC Layers
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5. Convolutional Neural Networks



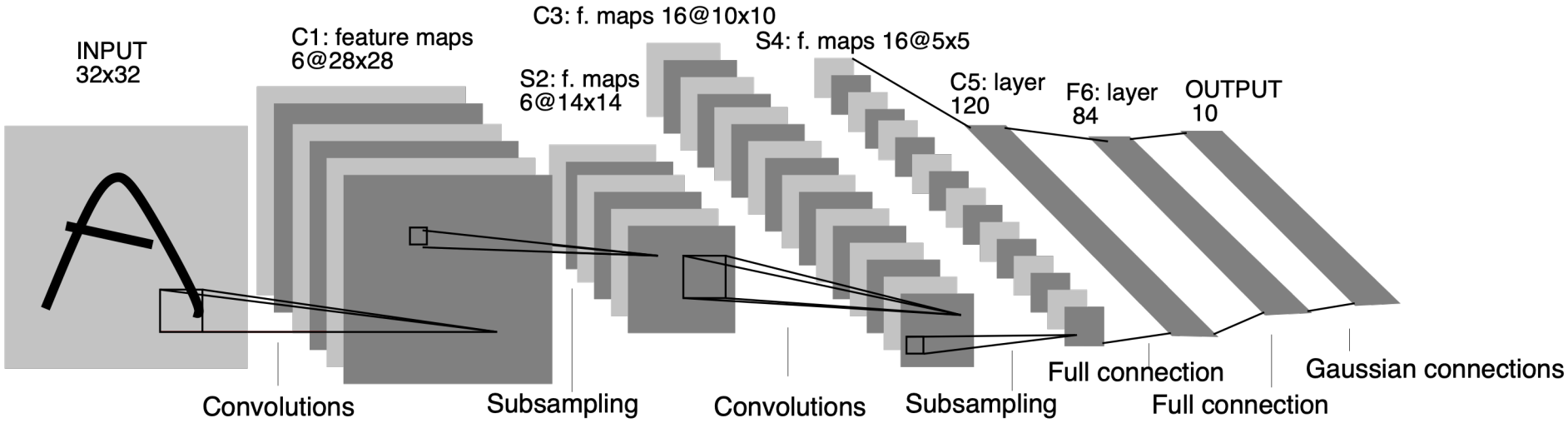
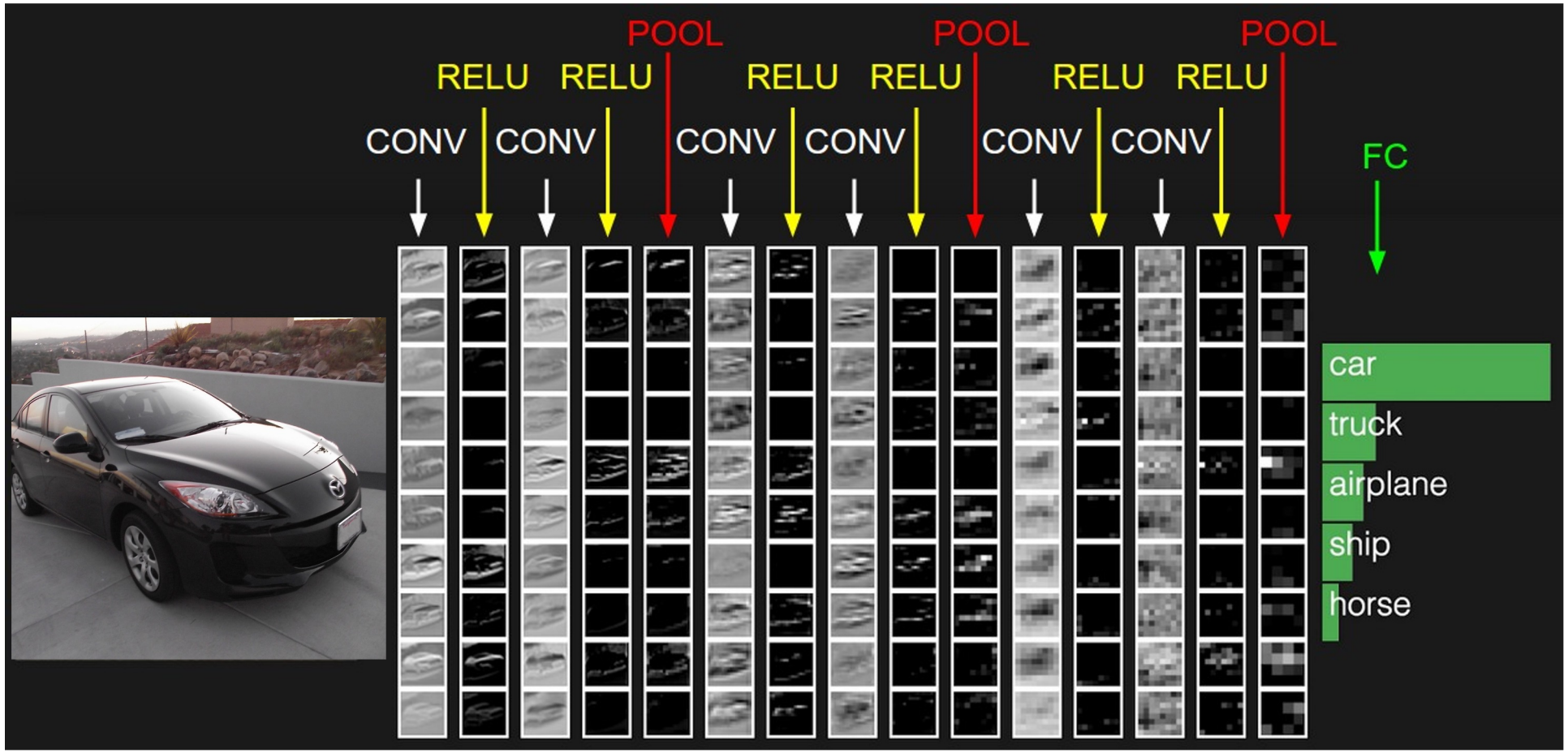


Image Credit: Yann LeCun, Kevin Murphy

preview:



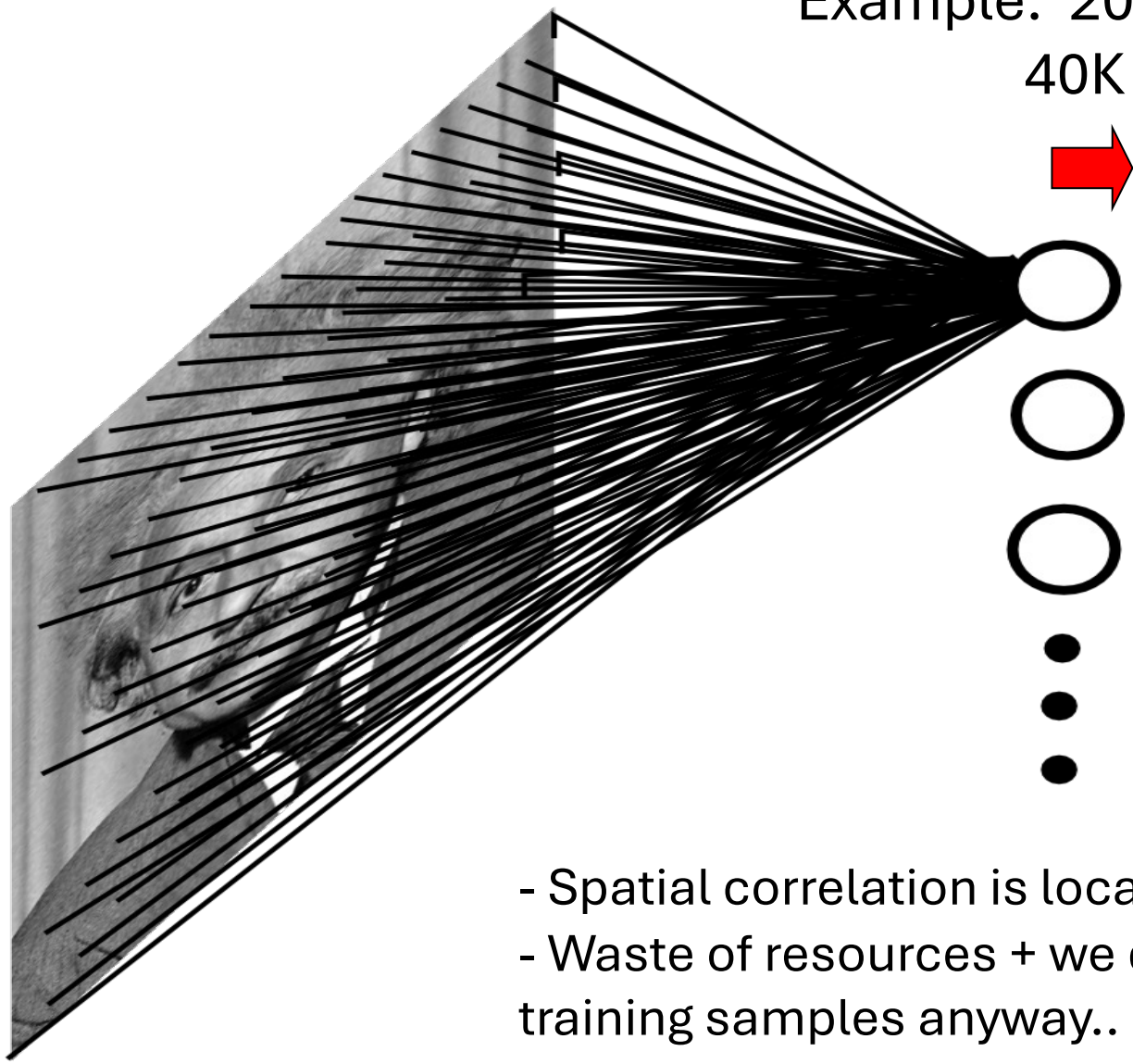
Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

Fully Connected Layer

Example: 200x200 image

40K hidden units

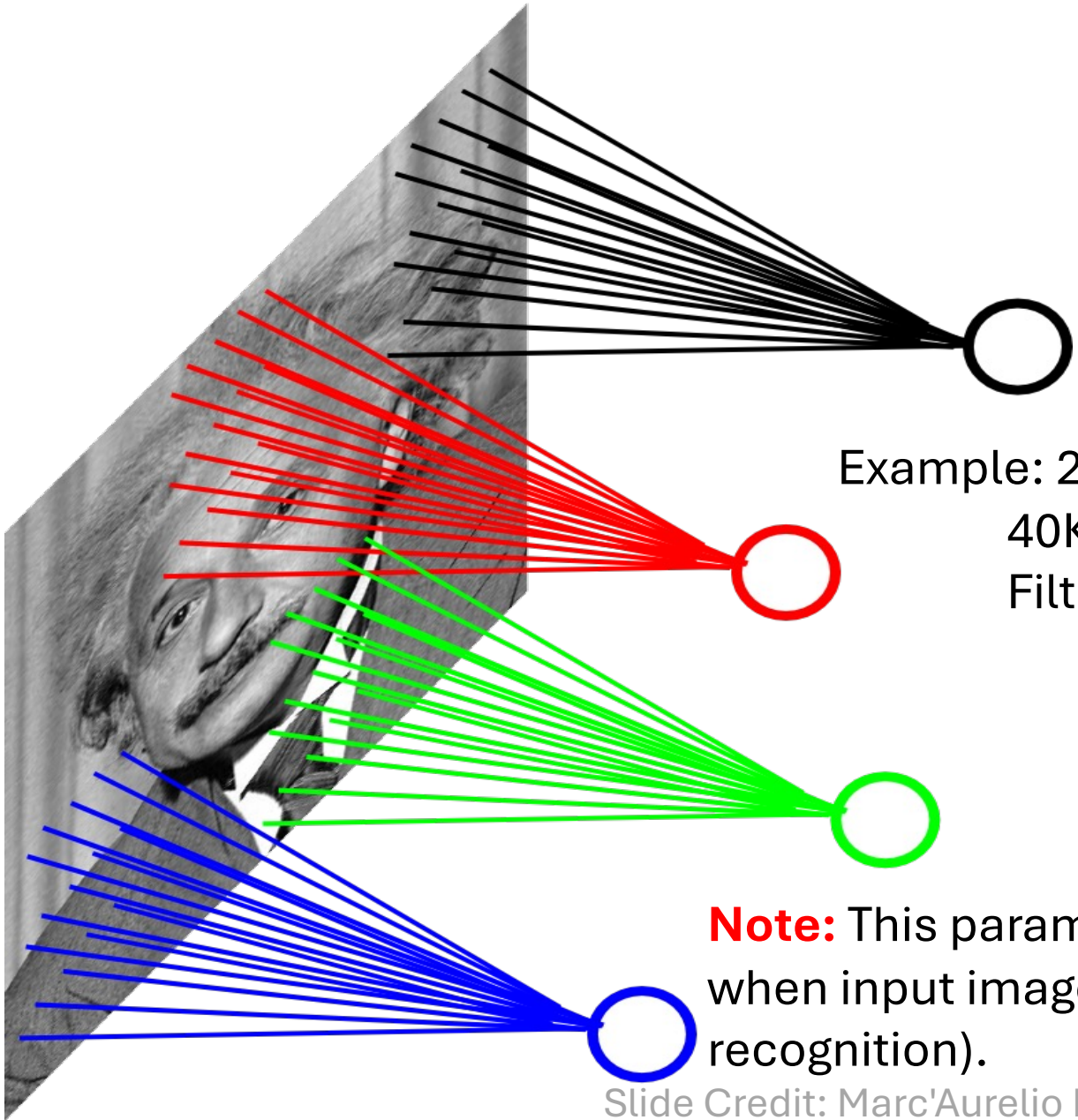
~1.6B parameters!!!



- Spatial correlation is local
- Waste of resources + we do not have enough training samples anyway..

Slide Credit: Marc'Aurelio Ranzato

Locally Connected Layer

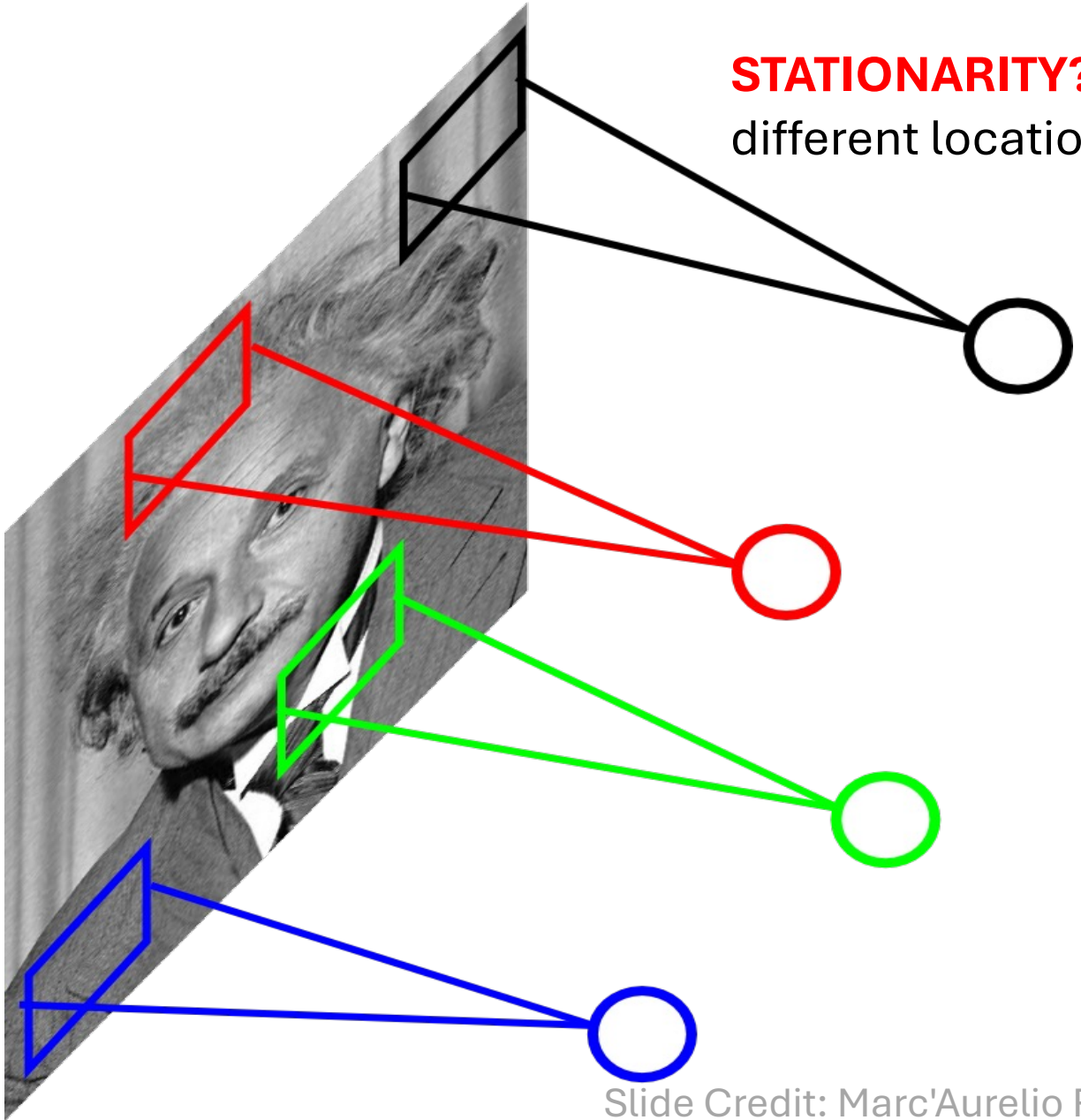


Example: 200x200 image
40K hidden units
Filter size: 10x10
4M parameters

Note: This parameterization is good when input image is registered (e.g., face recognition).

Slide Credit: Marc'Aurelio Ranzato

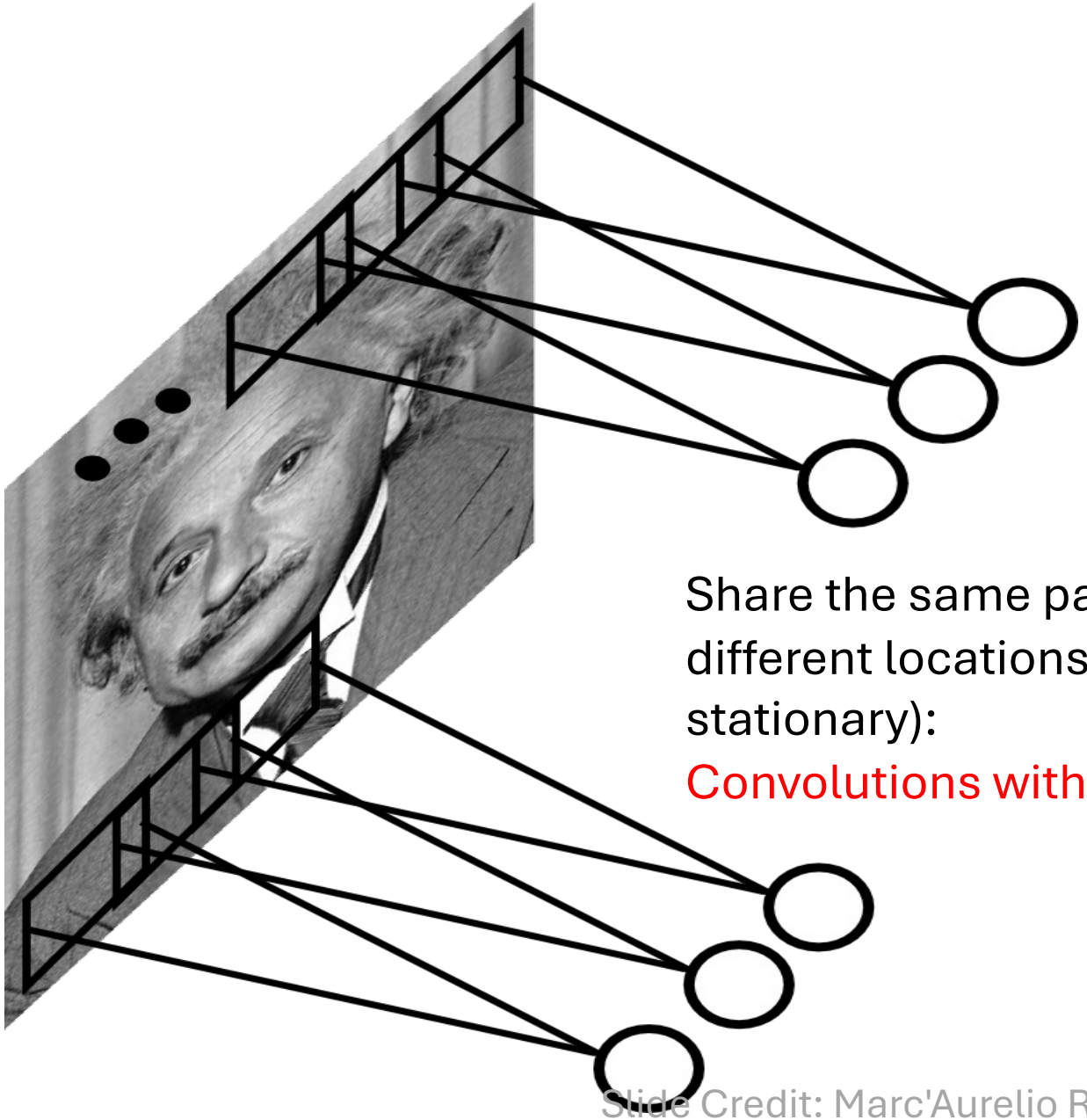
Locally Connected Layer



STATIONARITY? Statistics is similar at different locations

Slide Credit: Marc'Aurelio Ranzato

Convolutional Layer

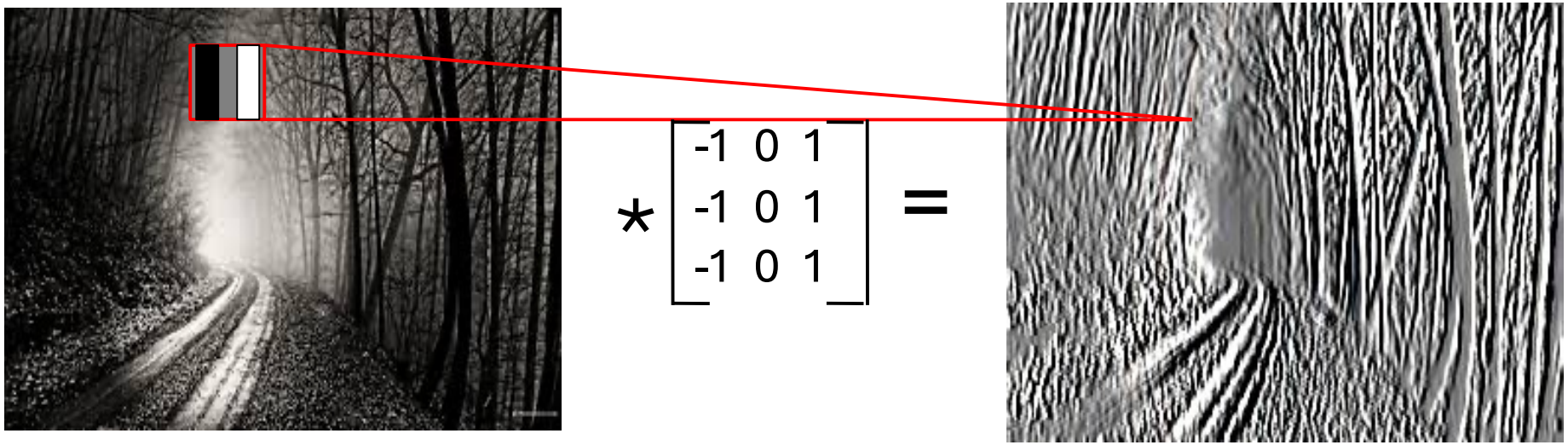


Share the same parameters across different locations (assuming input is stationary):

Convolutions with learned kernels

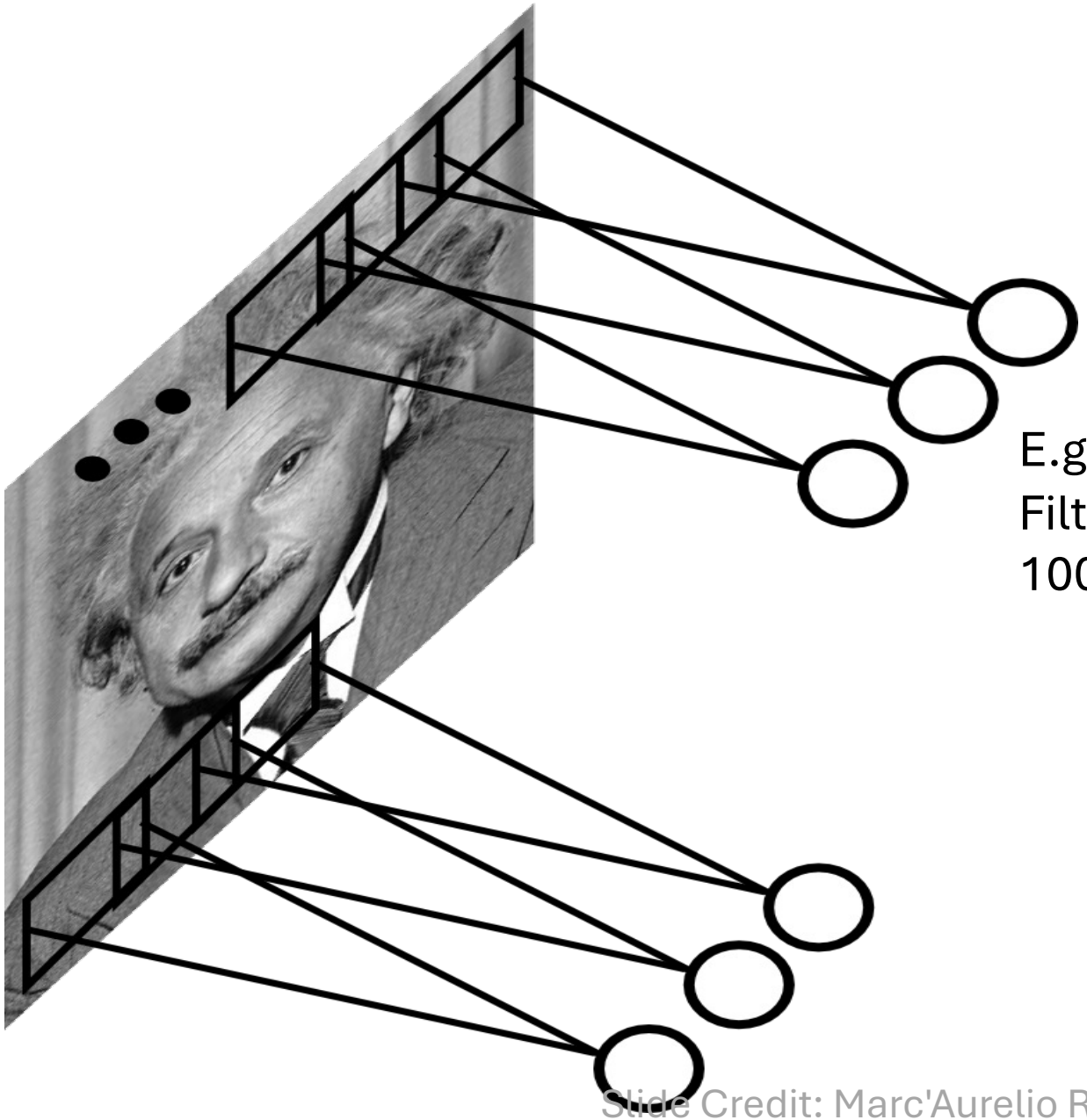
Slide Credit: Marc'Aurelio Ranzato

Convolutional Layer



Slide Credit: Marc'Aurelio Ranzato

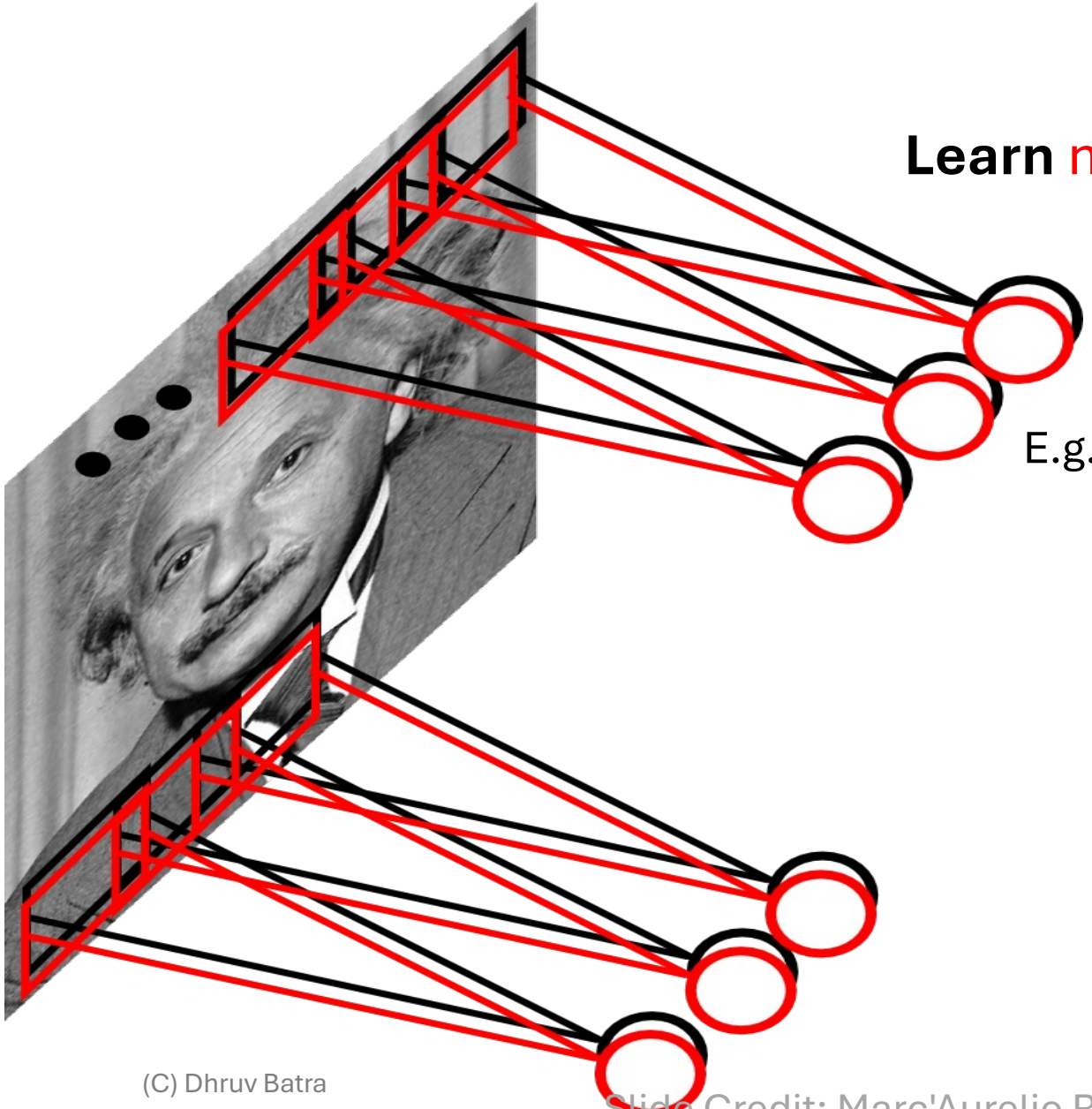
Convolutional Layer



E.g.: 200x200 image
Filter size: 10x10
100 parameters !!!!

Slide Credit: Marc'Aurelio Ranzato

Multi-channel Convolutional Layer



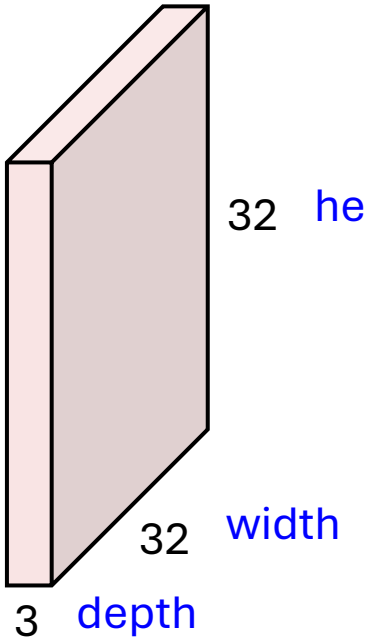
Learn multiple filters.

E.g.: 200x200 image
50 Filters
Filter size: 10x10
5K parameters: still tiny << 1.6B

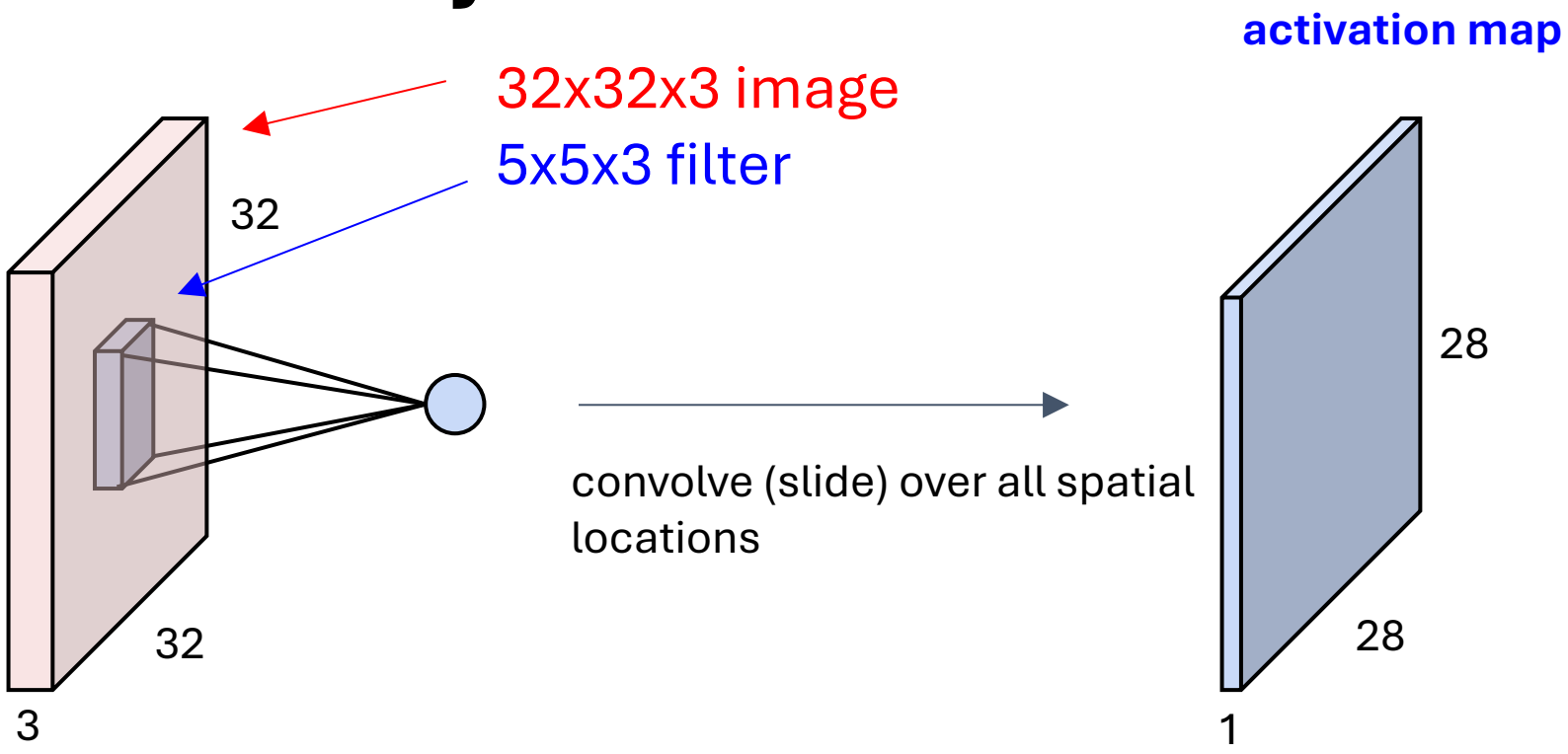
Slide Credit: Marc'Aurelio Ranzato

Convolution Layer

32x32x3 image -> preserve spatial structure



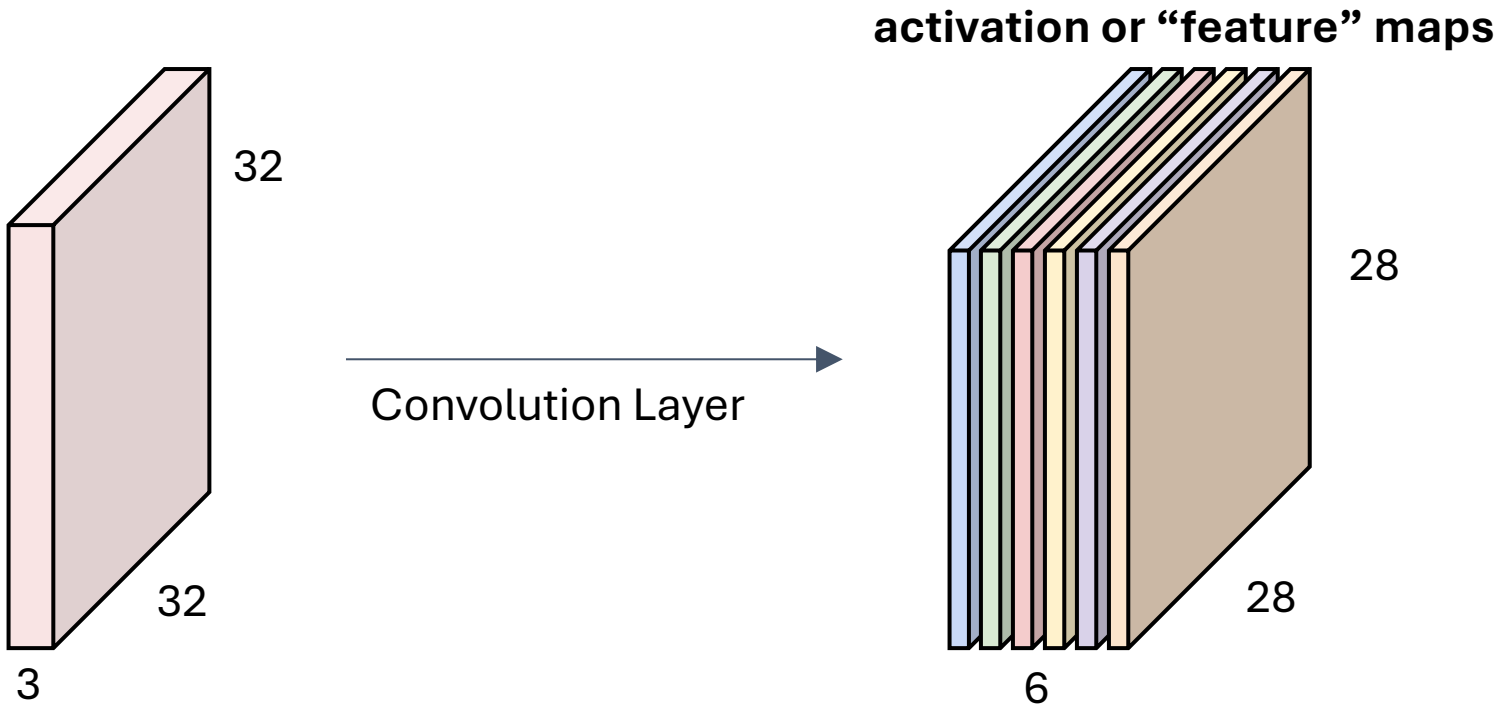
Convolution Layer



Number of parameters here?

5x5x3 = 75 !

Multiple filters: if we have 6 5x5 filters, we'll get 6 separate activation maps:

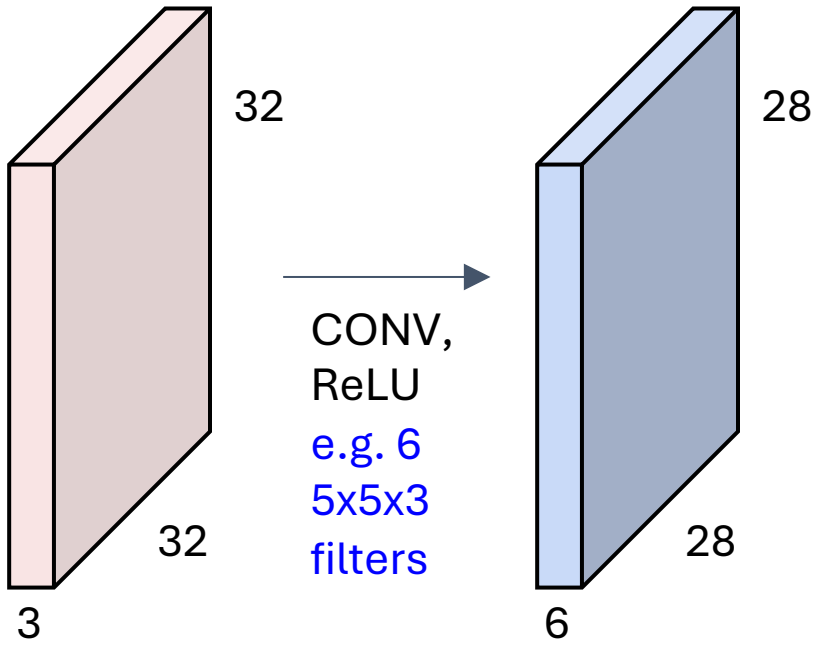


We stack these up to get a "new image" of size 28x28x6!

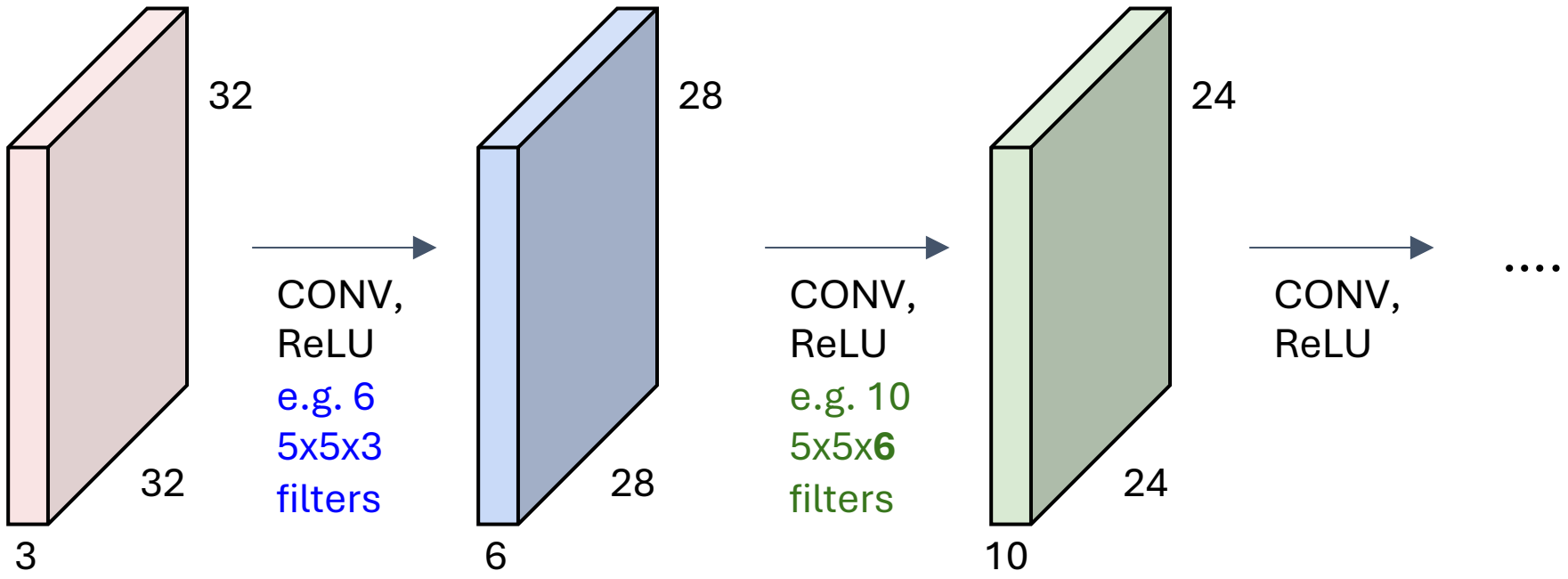
Number of parameters here?

6 filters * 5x5x3 = 6 x 75 = 450 !

Preview: ConvNet is a sequence of Convolution Layers, interspersed with activation functions

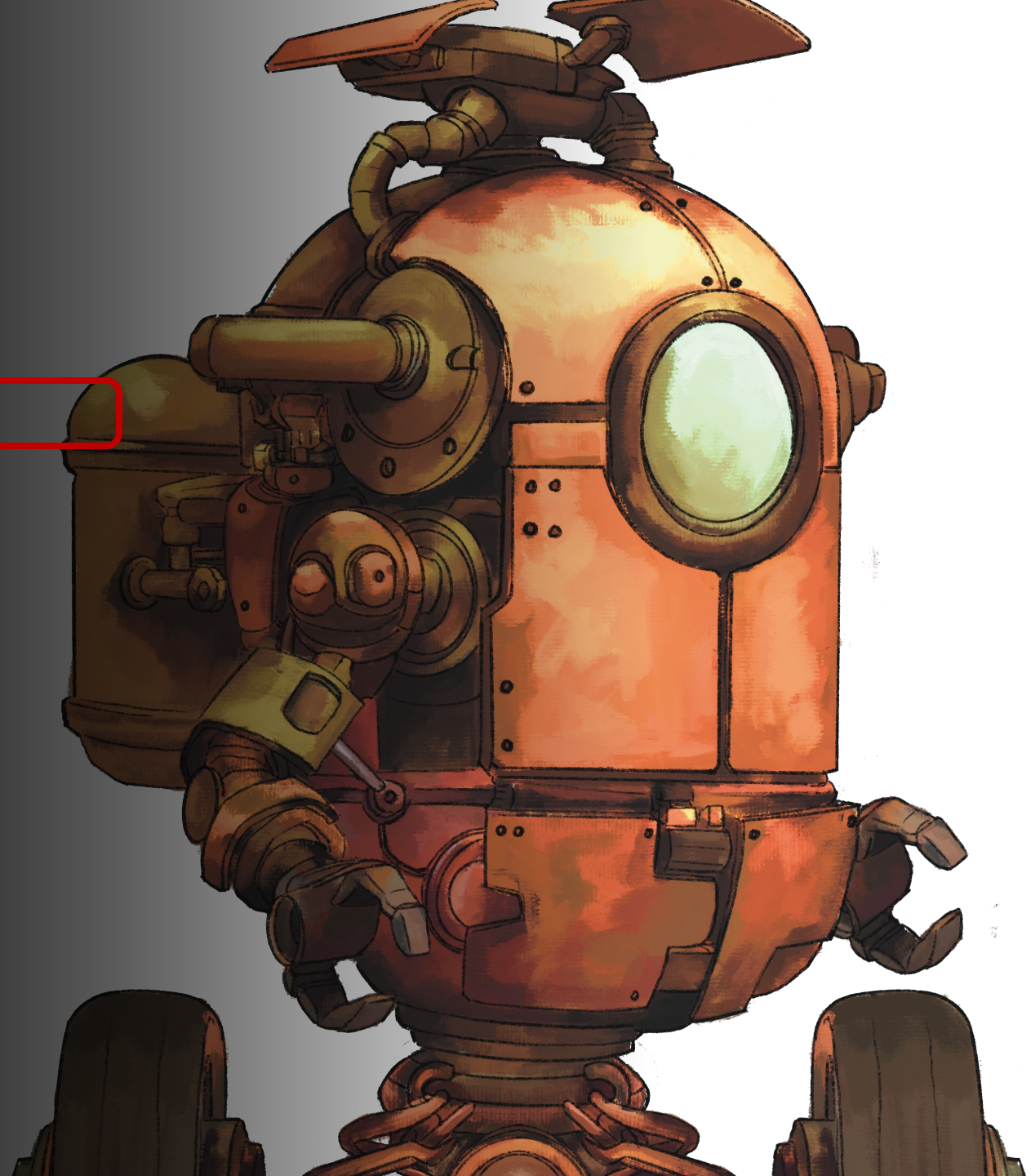


Preview: ConvNet is a sequence of Convolutional Layers, interspersed with **activation functions**



Outline

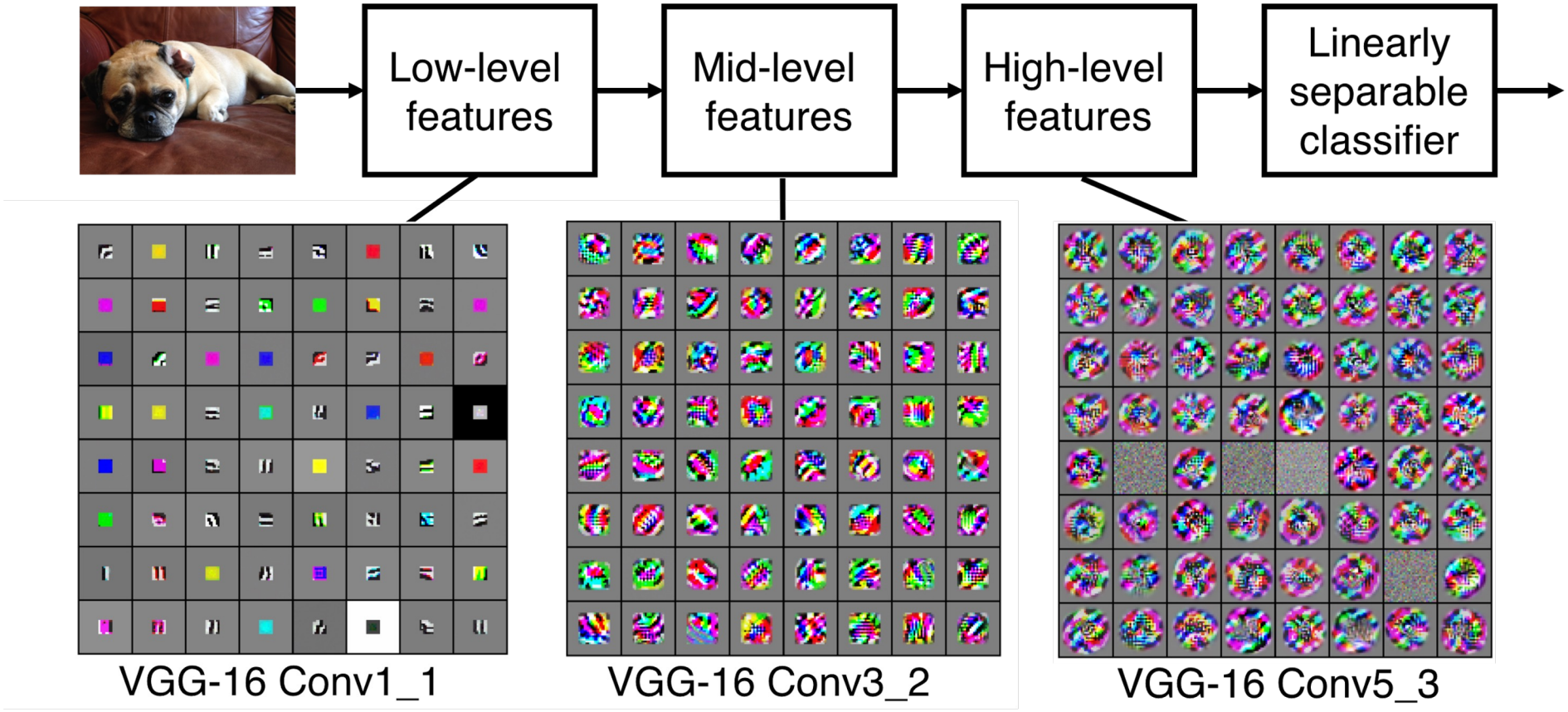
5. Convolutional Neural Nets
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6. Visualizing Learned Filters

[Zeiler and Fergus 2013]

Visualization of VGG-16 by Lane McIntosh. VGG-16 architecture from [Simonyan and Zisserman 2014].



Layers 1,2

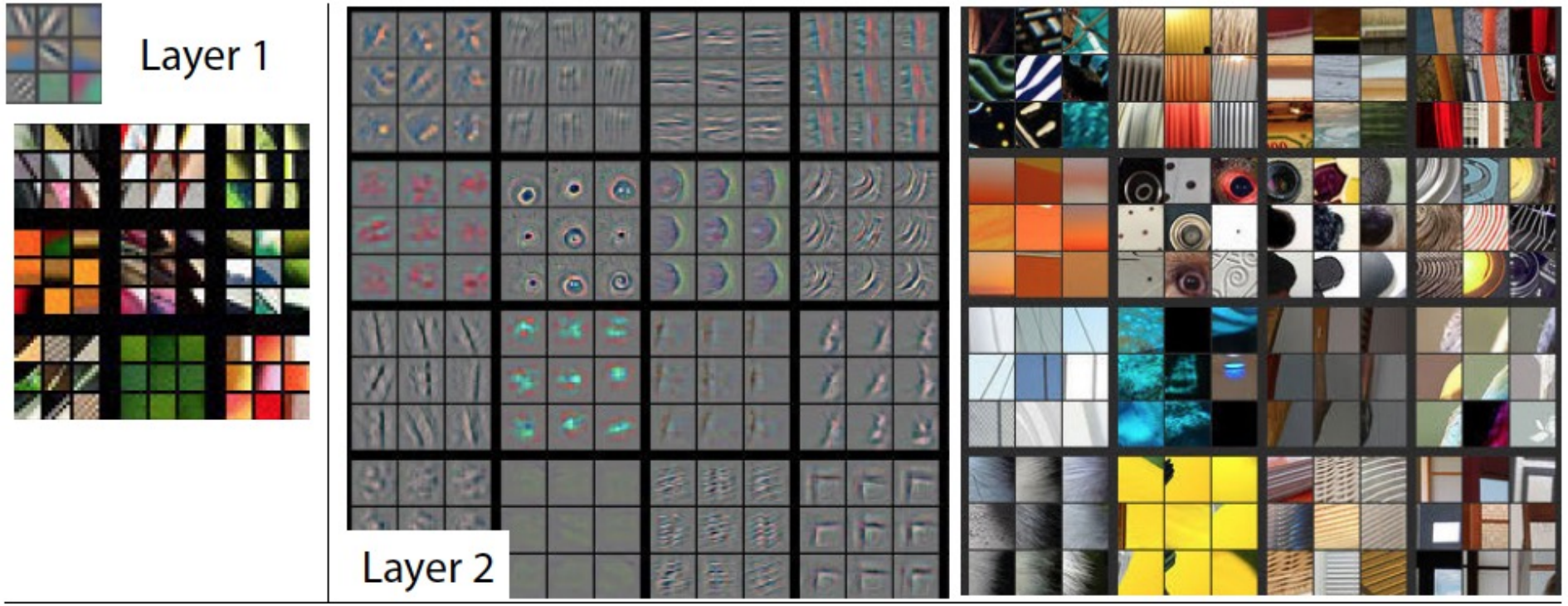


Figure Credit: [Zeiler & Fergus ECCV14]

Layer 3

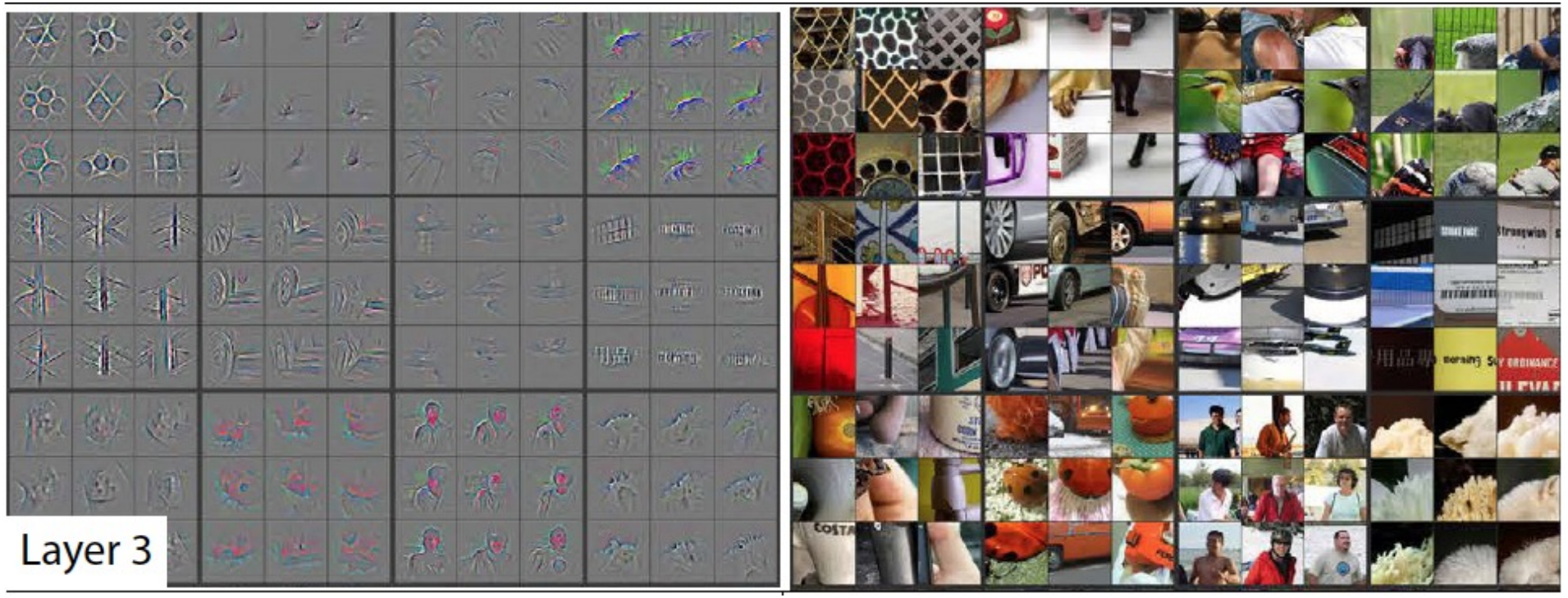


Figure Credit: [Zeiler & Fergus ECCV14]

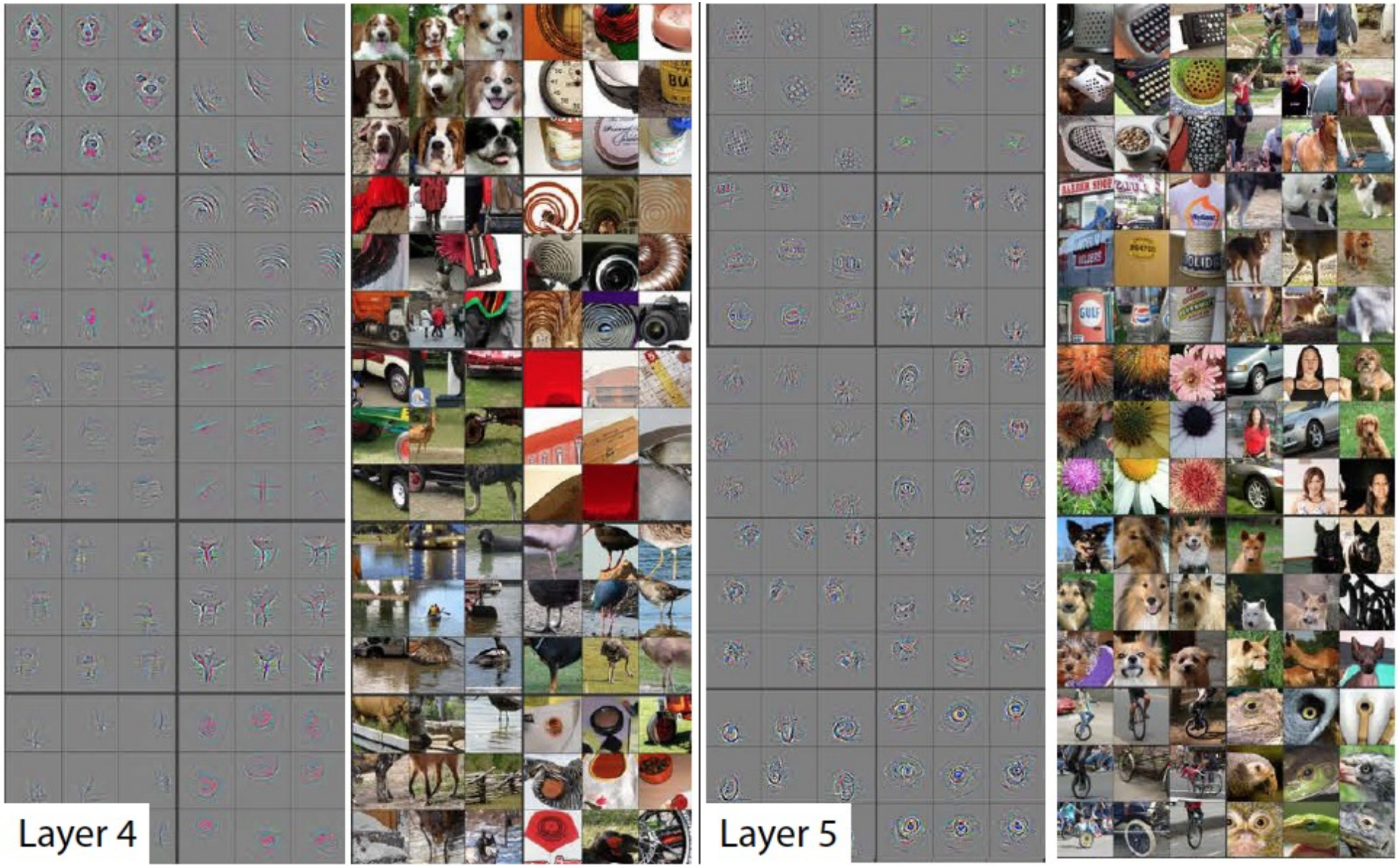
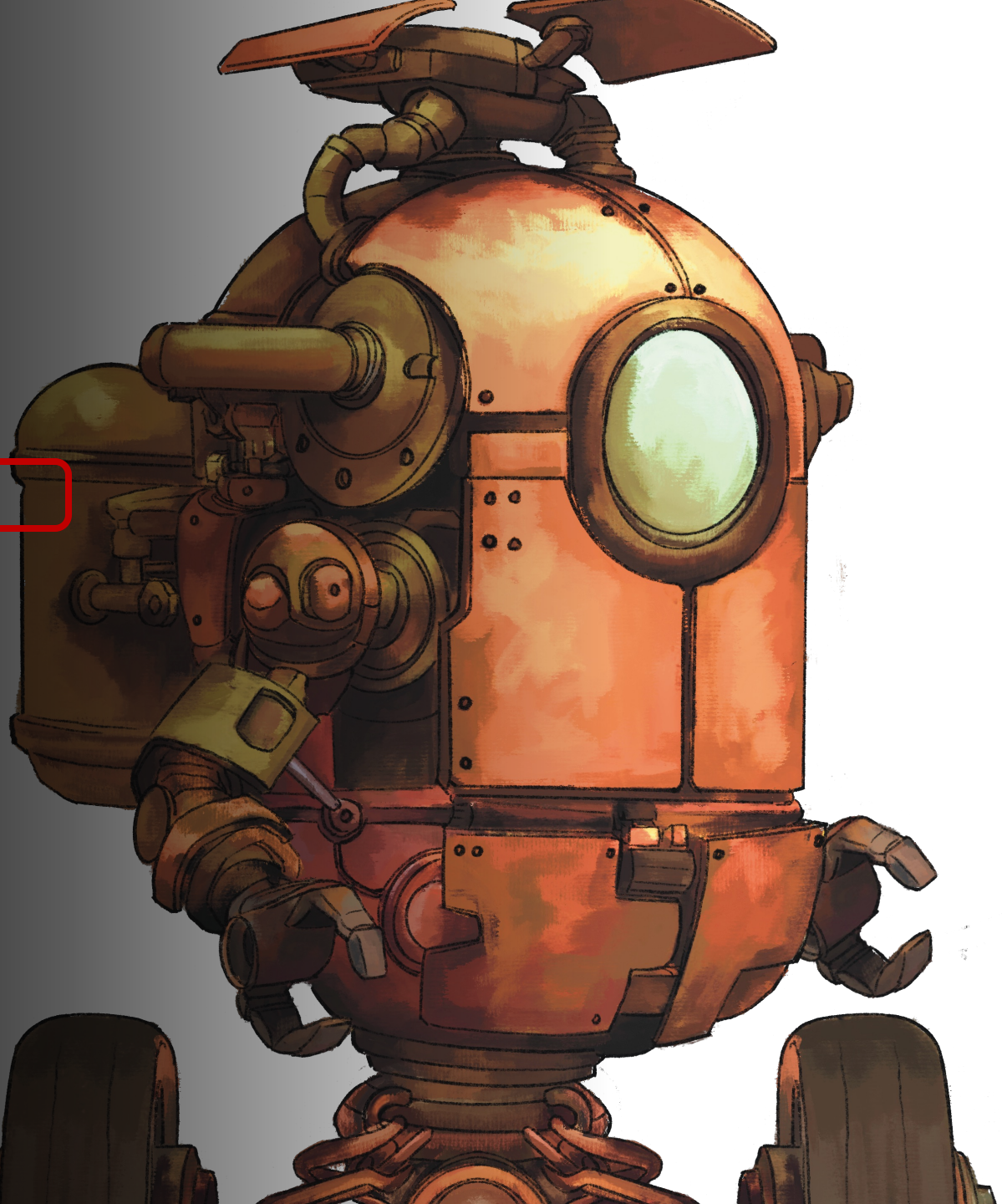


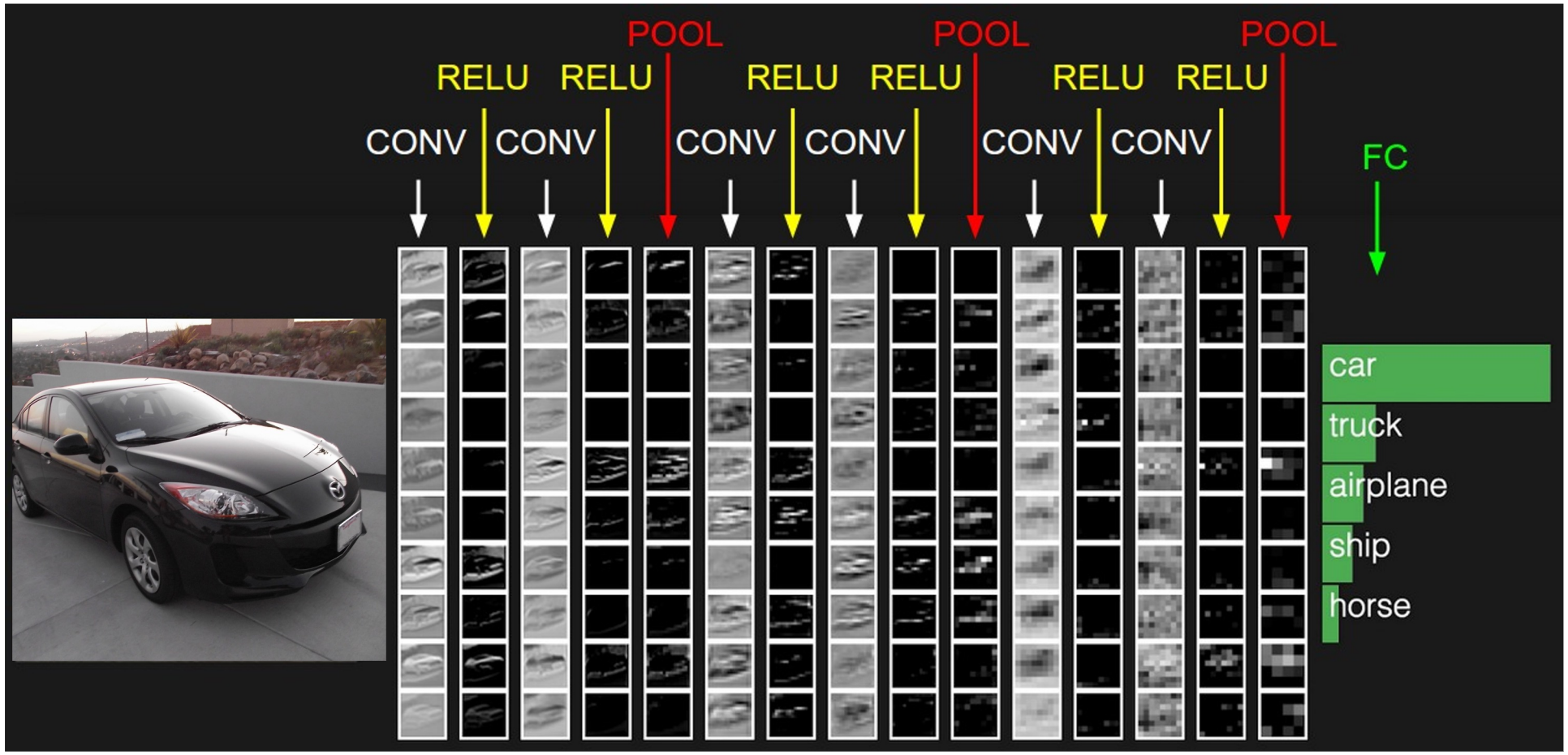
Figure Credit: [Zeiler & Fergus ECCV14]

Outline

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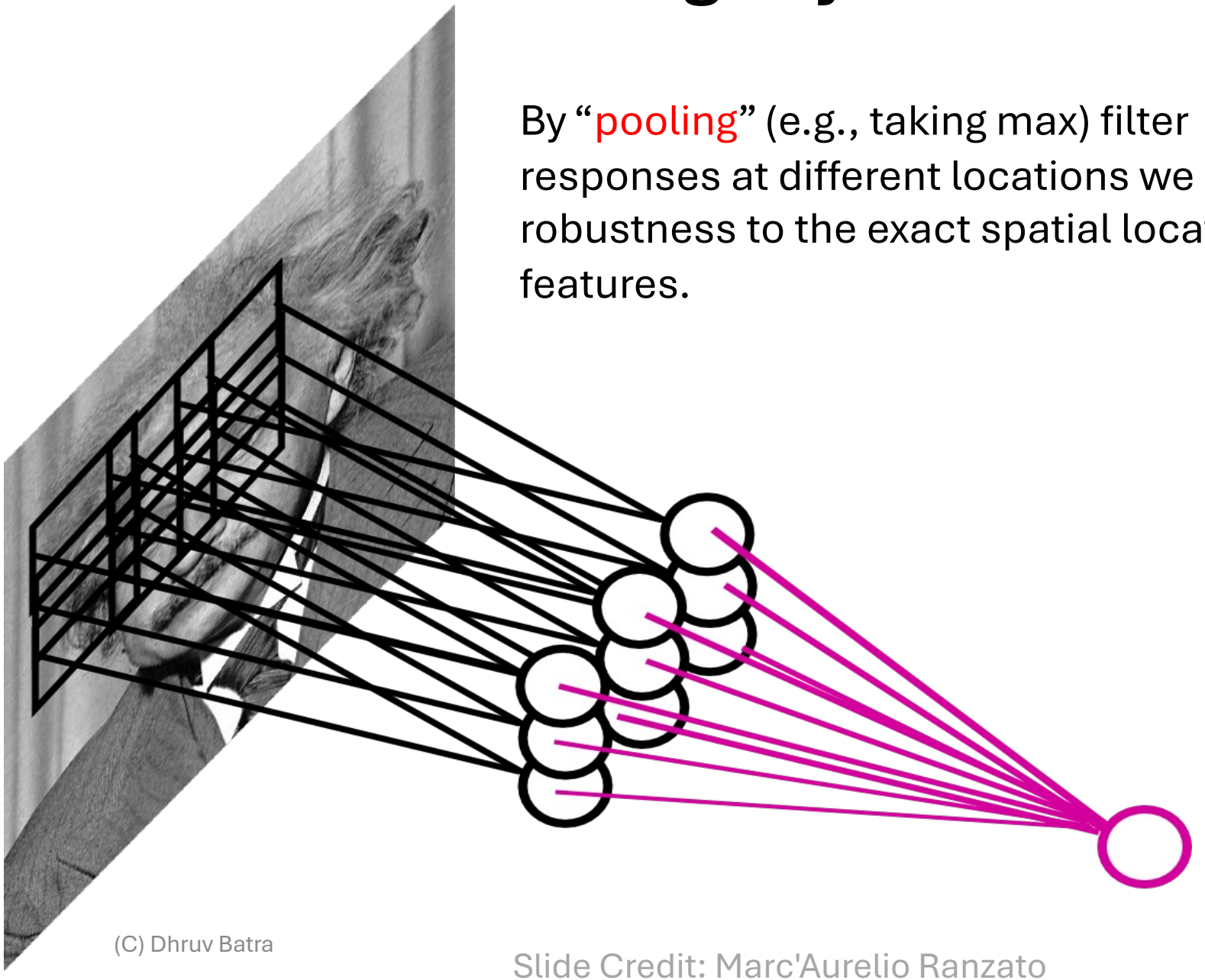


7. Pooling and Fully Convolutional Classifier



Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

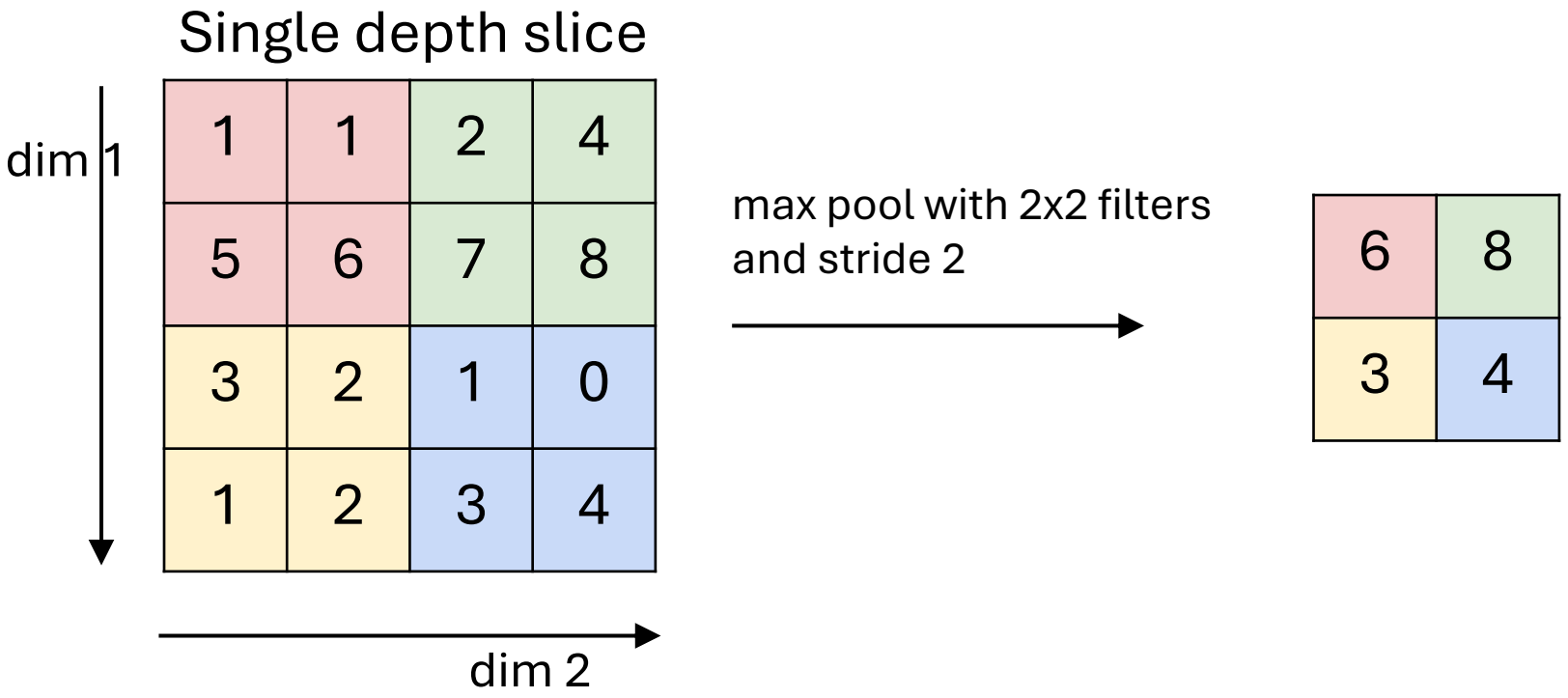
Pooling Layer



By “pooling” (e.g., taking max) filter responses at different locations we gain robustness to the exact spatial location of features.

Slide Credit: Marc'Aurelio Ranzato

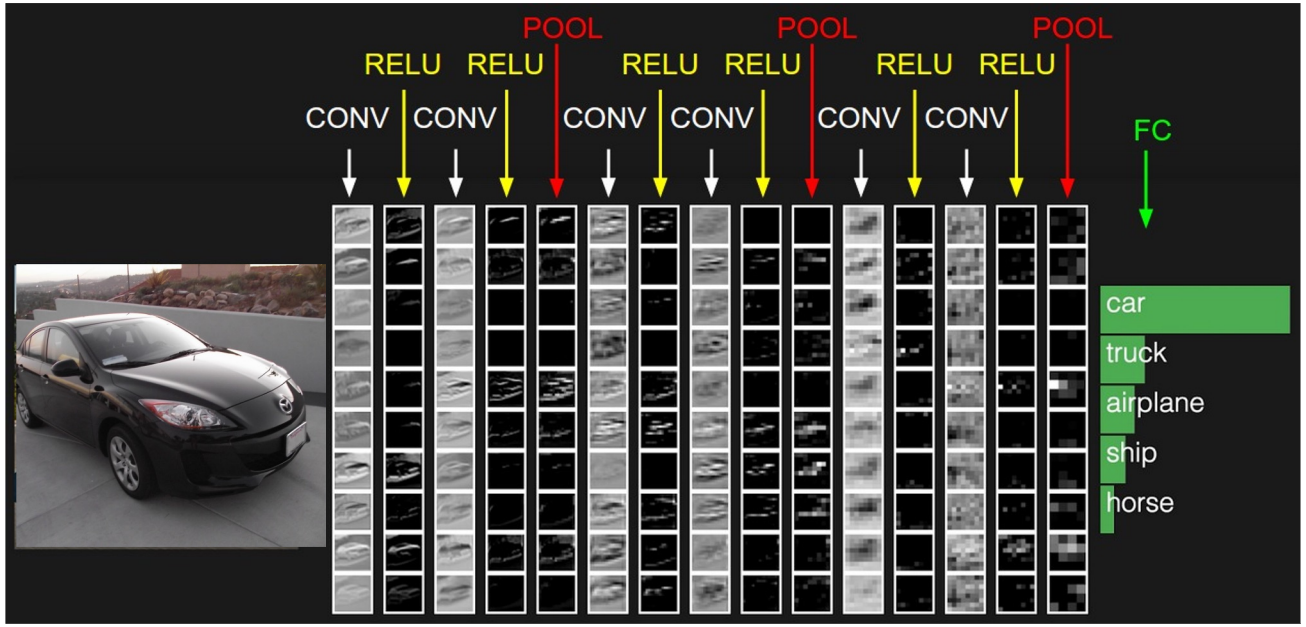
MAX POOLING



Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

Fully Connected Layer (FC layer)

- Contains neurons that connect to the entire input volume, as in multi-layer perceptrons



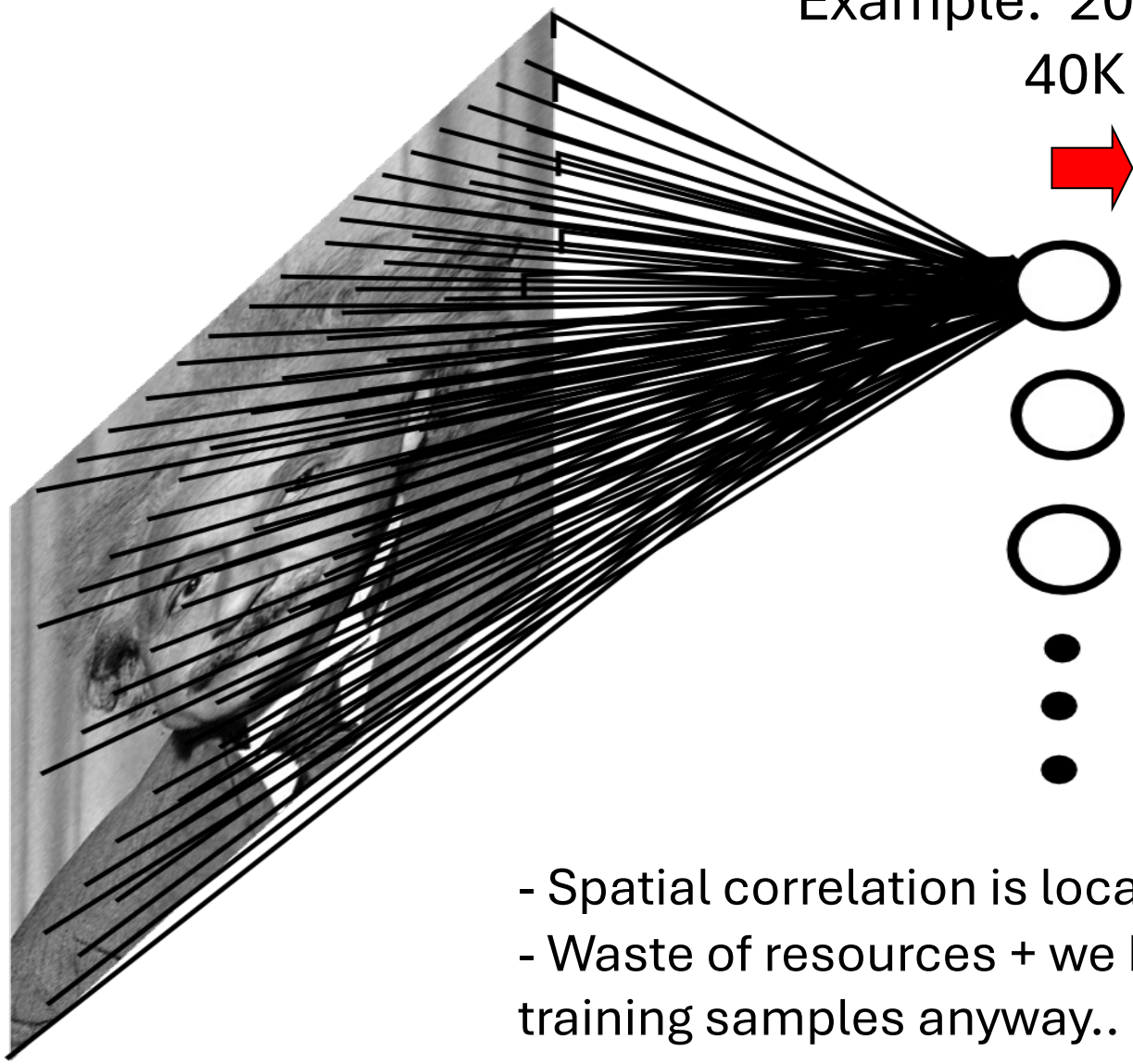
Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

Fully Connected Layer

Example: 200x200 image

40K hidden units

~2B parameters!!!

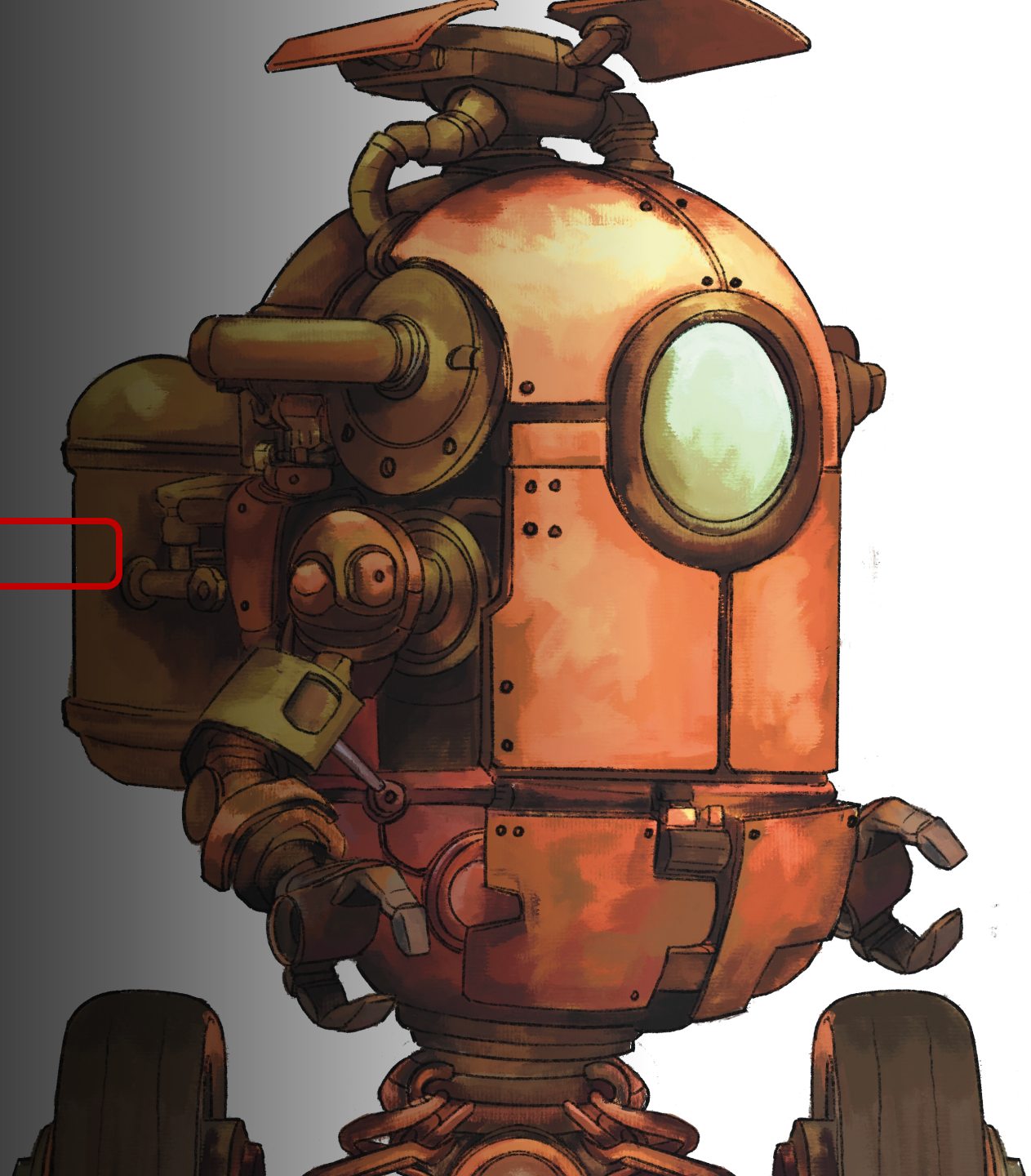


- Spatial correlation is local
- Waste of resources + we have not enough training samples anyway..

Slide Credit: Marc'Aurelio Ranzato

Outline

5. Convolutional Neural Nets
6. Visualizing Learned Filters
7. Pooling and FC Layers
8. Applications in Vision
9. Object Classification
10. Applications in Robotics



8. Applications in Vision

- From pixels to concepts:
  - Image processing
  - Object classification
  - Object detection
  - Pixelwise segmentation

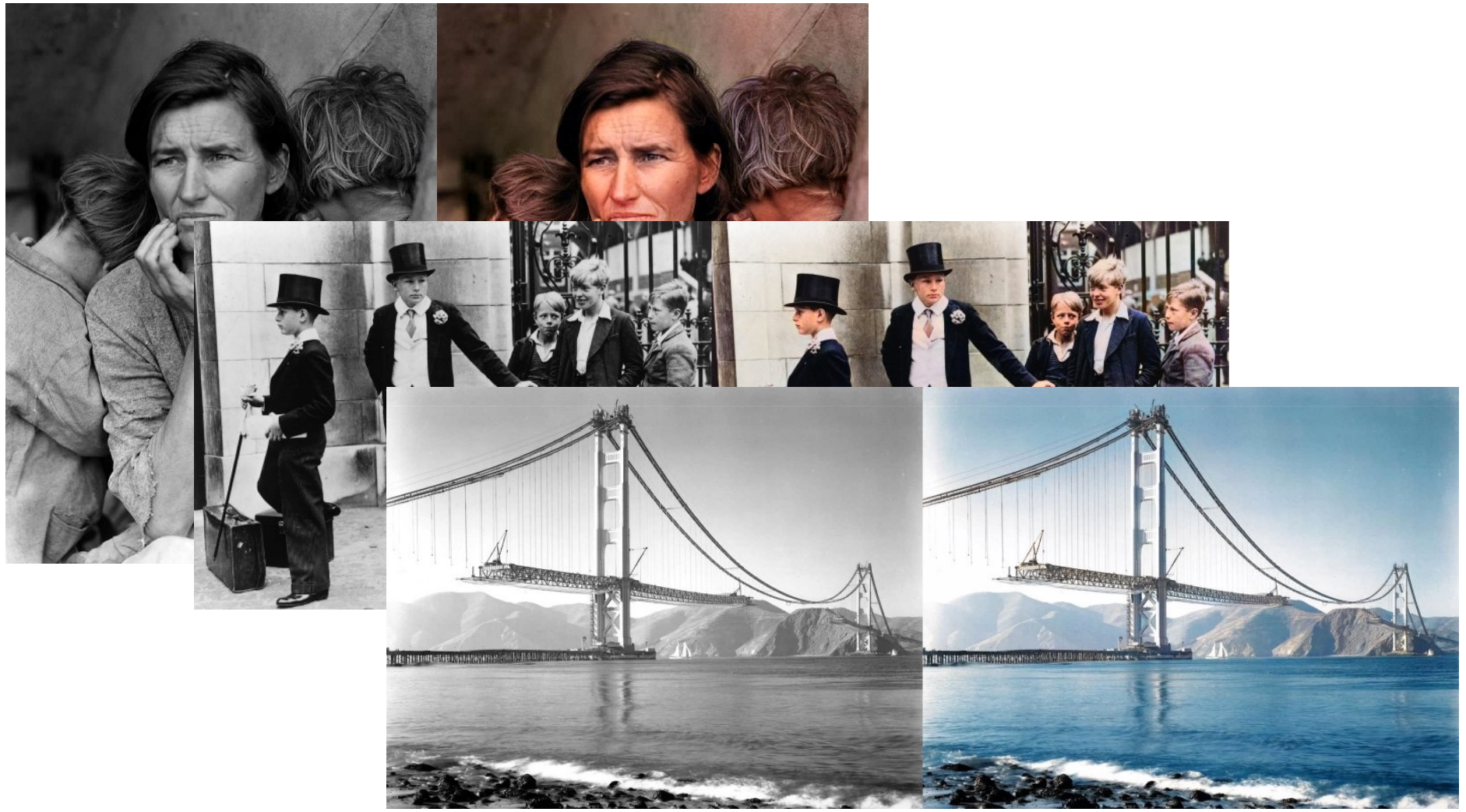
Colorization

- Given a grayscale image, colorize the image realistically
- Zhang et al. pose colorization as classification task and use class-rebalancing to improve results
- Demonstrate higher rates of fooling humans using “colorization Turing test”



Colorful Image Colorization. Richard Zhang, Phillip Isola, Alexei A. Efros. ECCV 2016.

DeOldify



https://github.com/jantic/DeOldify

Super-Resolution

Low resolution





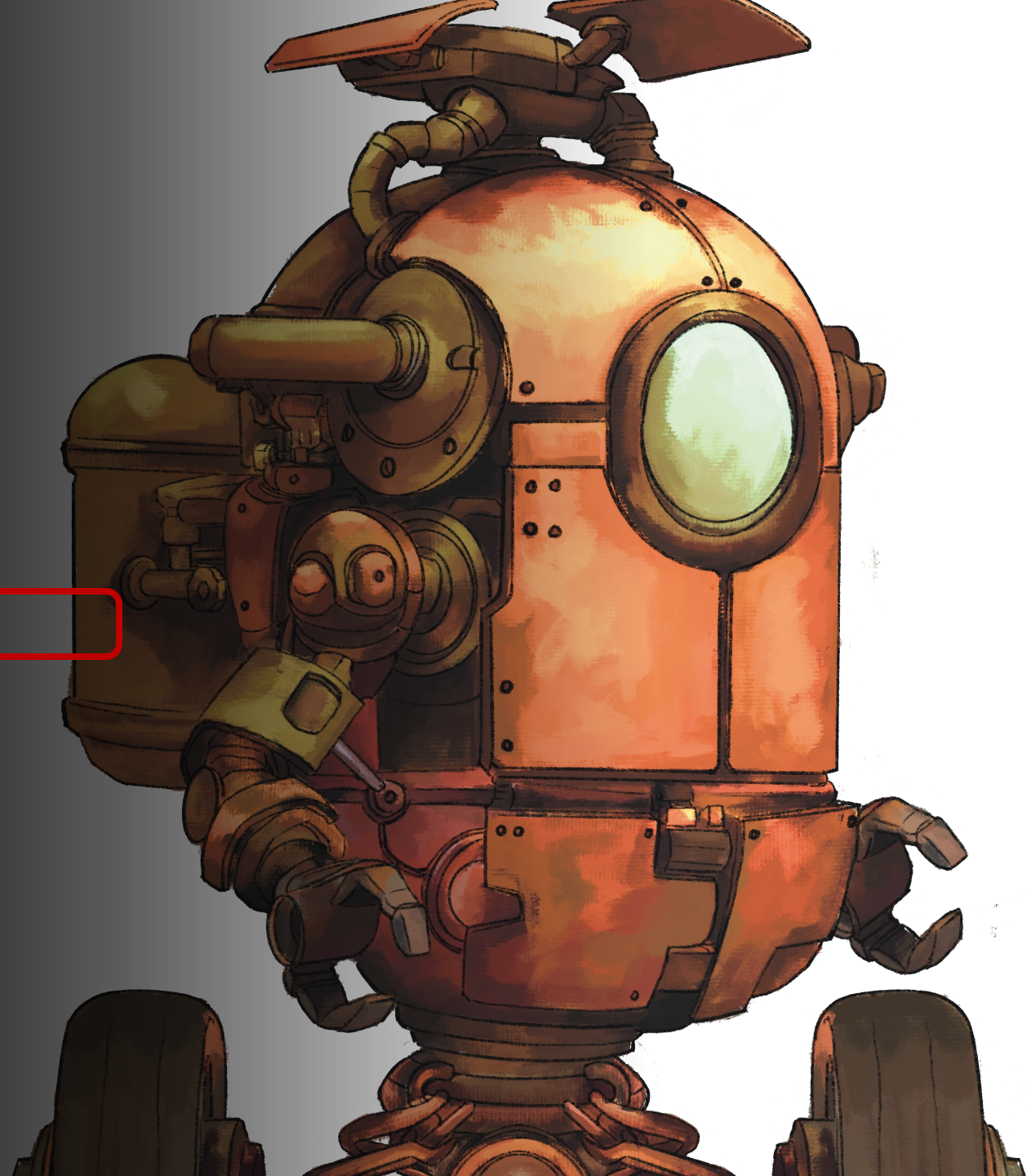
High resolution





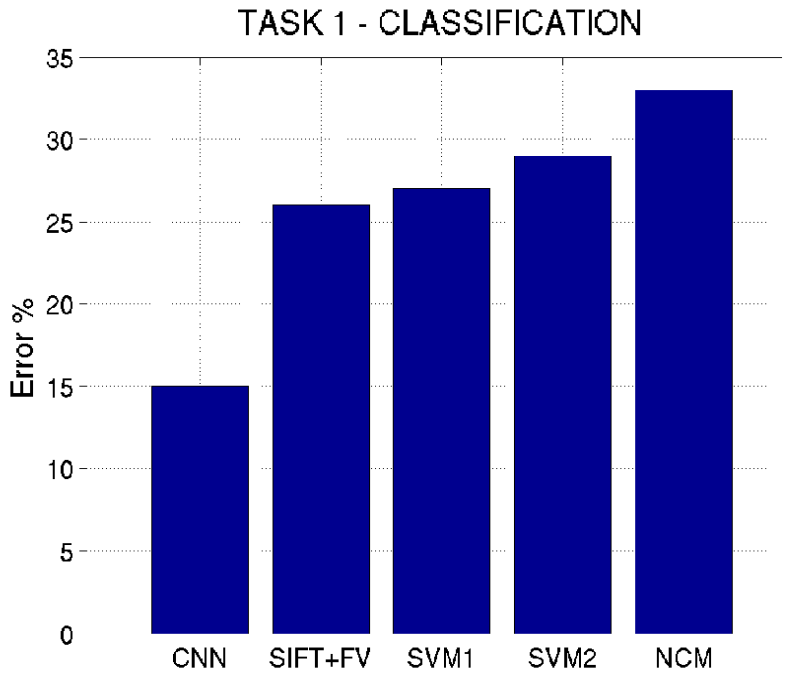
Outline

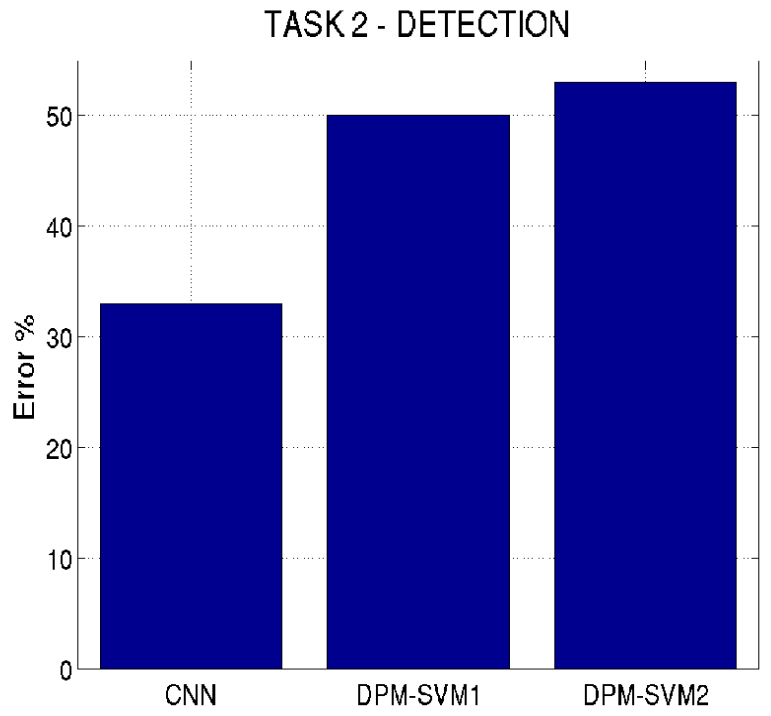
5. Convolutional Neural Nets
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9. Object Classification Revolution

Results: ILSVRC 2012





Krizhevsky et al. "ImageNet Classification with deep CNNs" NIPS 2012



mite

mite
black widow
cockroach
tick
starfish



container ship

container ship
lifeboat
amphibian
fireboat
drilling platform



motor scooter

motor scooter
go-kart
moped
bumper car
golfcart



leopard

leopard
jaguar
cheetah
snow leopard
Egyptian cat



grille

convertible
grille
pickup
beach wagon
fire engine



mushroom

agaric
mushroom
jelly fungus
gill fungus
dead-man's-fingers



cherry

dalmatian
grape
elderberry
ffordshire bullterrier
currant

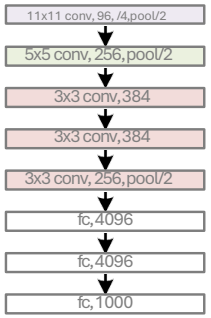


Madagascar cat

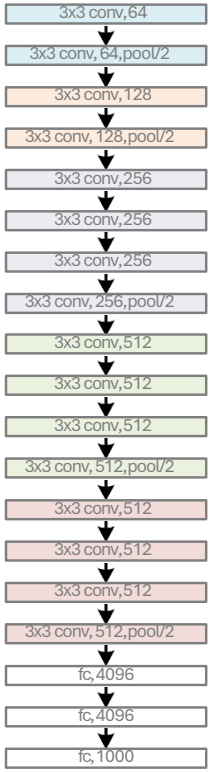
squirrel monkey
spider monkey
titi
indri
howler monkey

Revolution of Depth

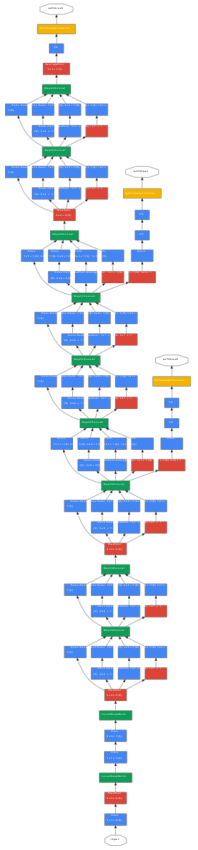
AlexNet, 8 layers
(ILSVRC 2012)



VGG, 19 layers
(ILSVRC 2014)



GoogleNet, 22 layers
(ILSVRC 2014)

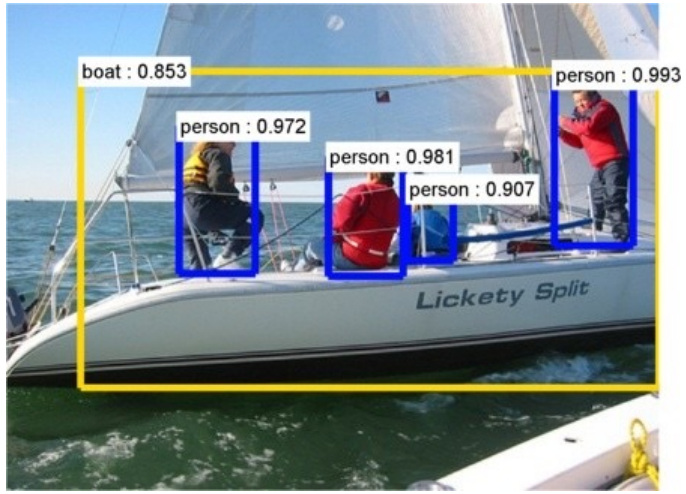


Kaiming He, Xiangyu Zhang, Shaoqing Ren, & Jian Sun. "Deep Residual Learning for Image Recognition". CVPR 2016.

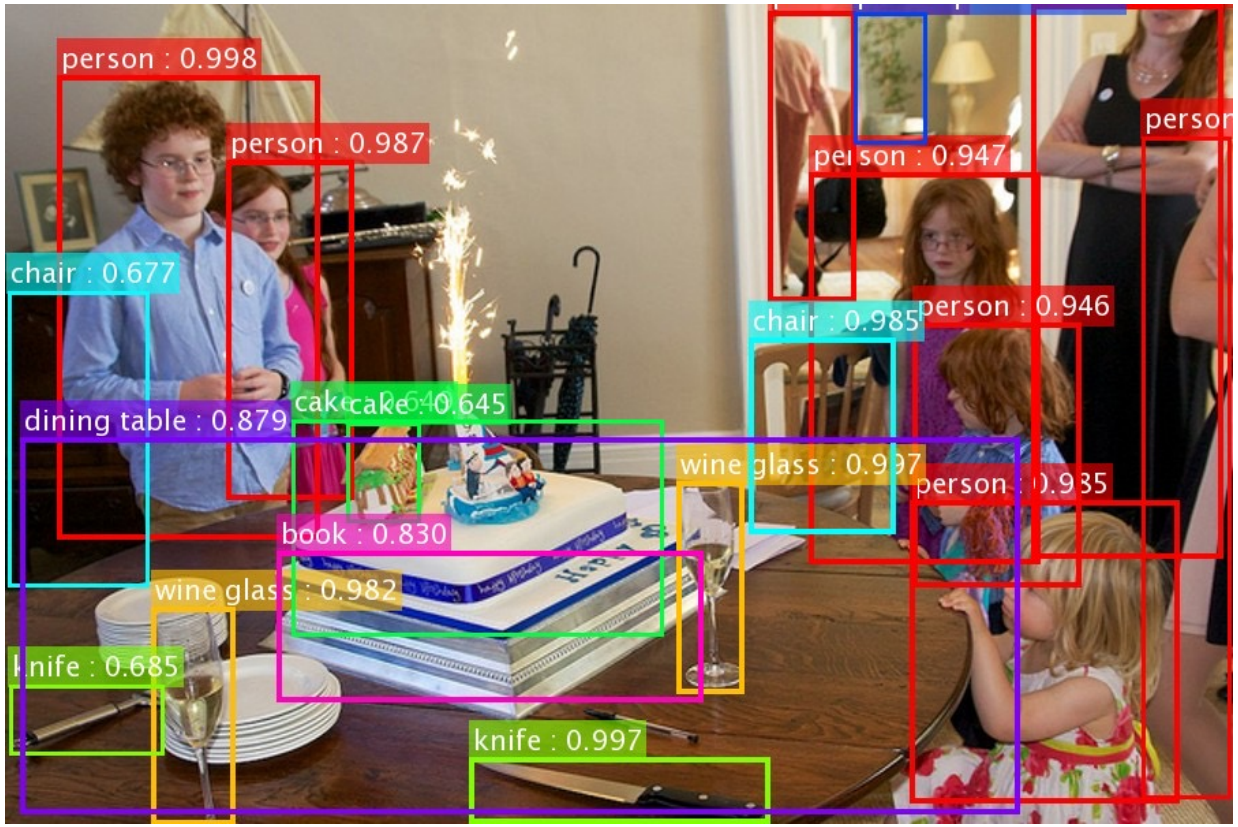
Object Detection



Image Classification
(what?)



Object Detection
(what + where?)



ResNet's object detection result on COCO

Kaiming He, Xiangyu Zhang, Shaoqing Ren, & Jian Sun. "Deep Residual Learning for Image Recognition". CVPR 2016.
Shaoqing Ren, Kaiming He, Ross Girshick, & Jian Sun. "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks". NIPS 2015.

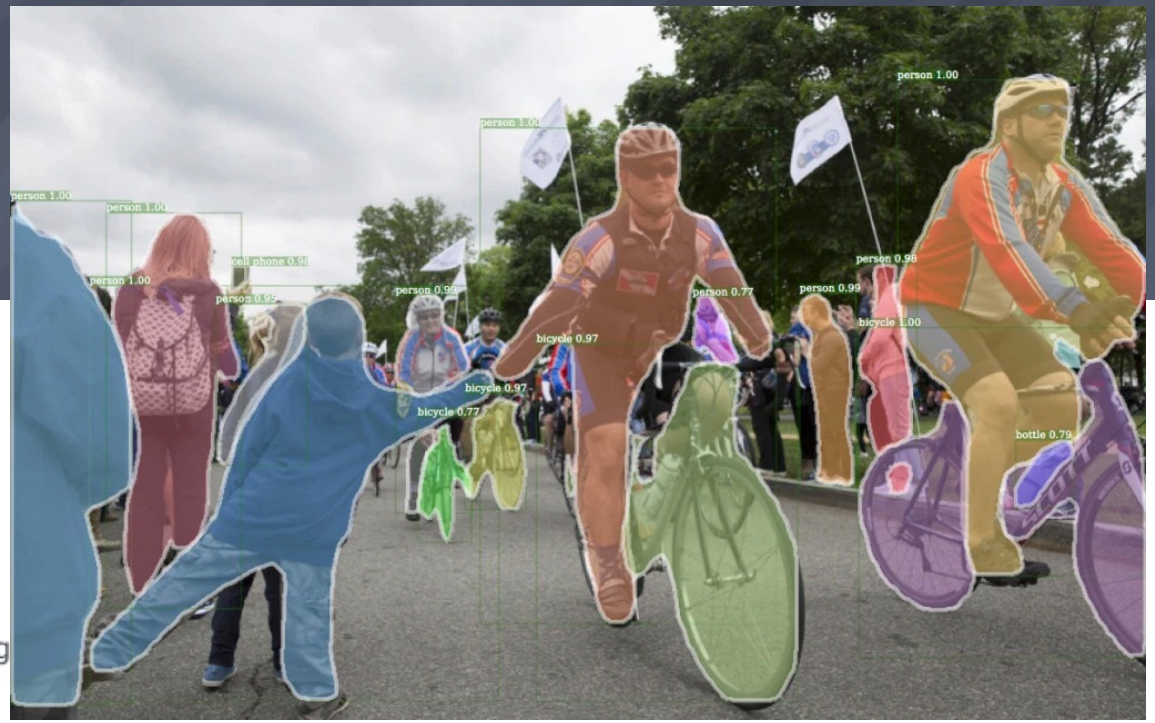


this video is available online: https://youtu.be/WZmSMkK9VuA

Results on real video. Models trained on MS COCO (80 categories). (frame-by-frame; no temporal processing)

Kaiming He, Xiangyu Zhang, Shaoqing Ren, & Jian Sun. "Deep Residual Learning for Image Recognition". CVPR 2016.
Shaoqing Ren, Kaiming He, Ross Girshick, & Jian Sun. "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks". NIPS 2015.

Detectron



Detectron includes implementations of the following

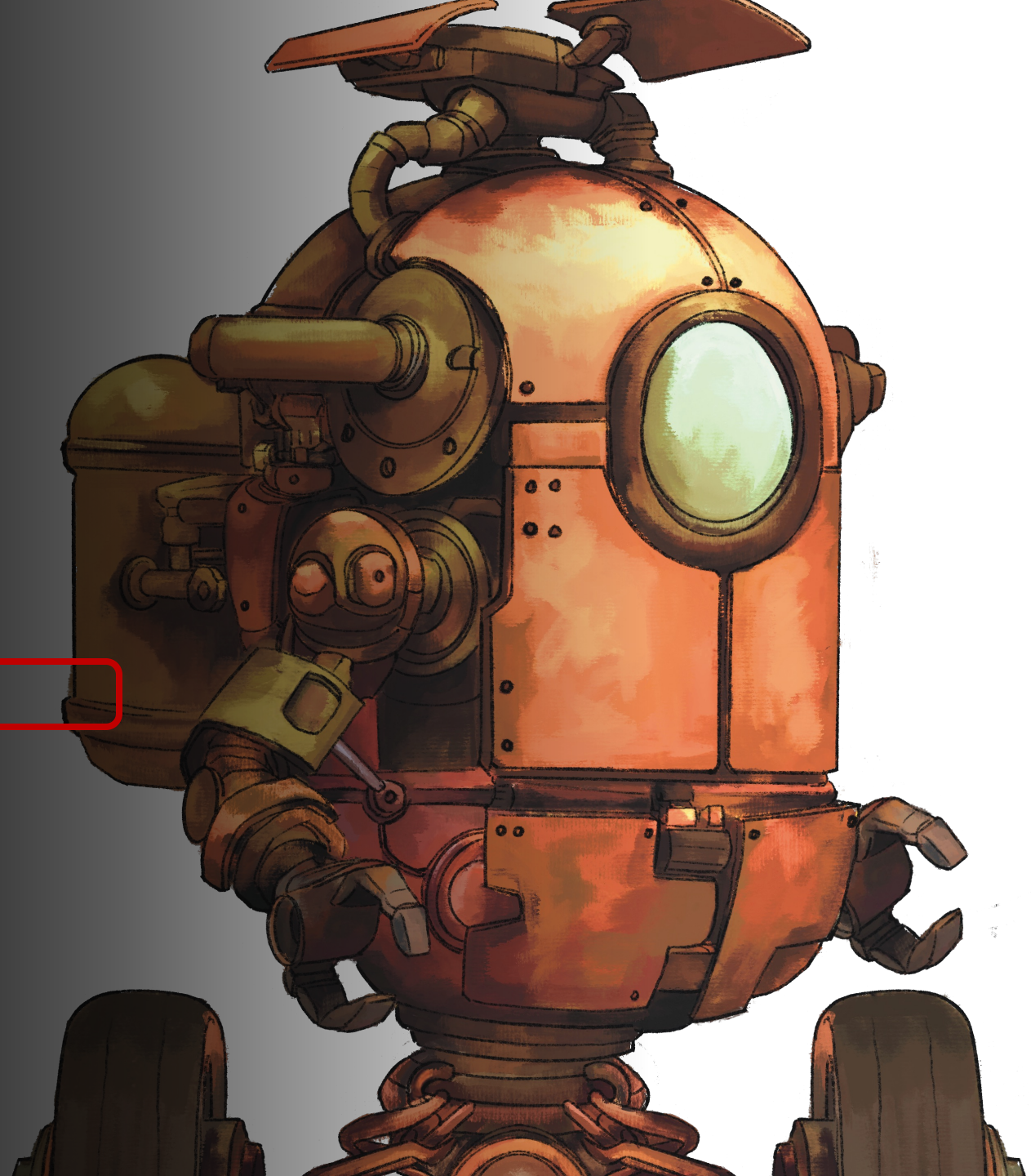
- Mask R-CNN — *Marr Prize at ICCV 2017*
- RetinaNet — *Best Student Paper Award at ICCV 2017*
- Faster R-CNN
- RPN
- Fast R-CNN
- R-FCN

using the following backbone network architectures:

- ResNeXt{50,101,152}
- ResNet{50,101,152}
- Feature Pyramid Networks (with ResNet/ResNeXt)
- VGG16

Outline

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10. Applications in Robotics

- From pixels to action:
  - Object Recognition
  - Depth from a Single Image
  - Deep Stereo
  - 3D Foundation Models

- Deep Reinforcement Learning: for another day!



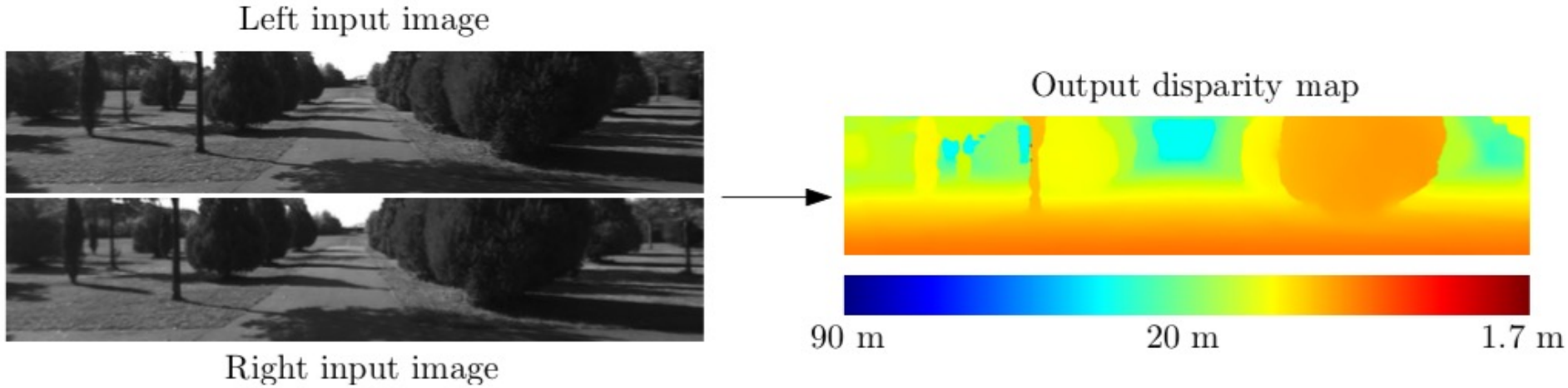
Verdant Robotics: Don’t’ shoot the Broccoli !

Deep Stereo

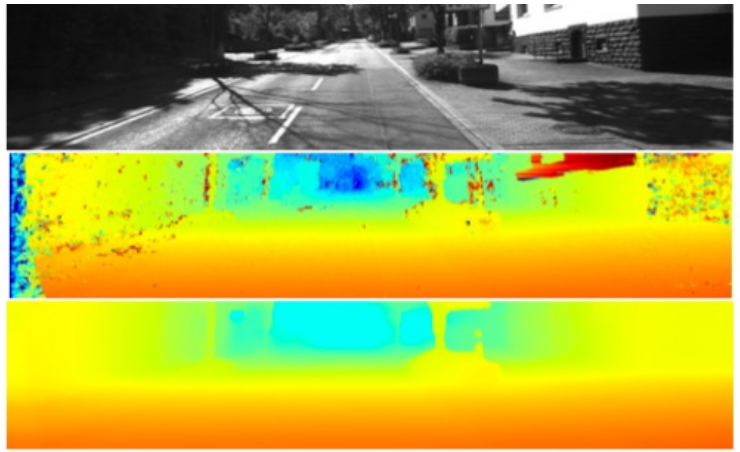
Computing the Stereo Matching Cost with a Convolutional Neural Network

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- Learns cost function



Winner-take all

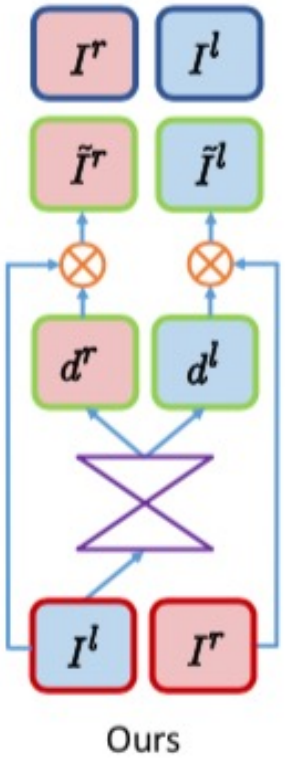
Smoothed

Monocular Depth

- Can we learn depth from a *single* image?
- Train on stereo, but test on mono!
- Learn to warp left to right and vice versa

Unsupervised Monocular Depth Estimation with Left-Right Consistency

Clément Godard Oisin Mac Aodha Gabriel J. Brostow
University College London
http://visual.cs.ucl.ac.uk/pubs/monoDepth/



Ours

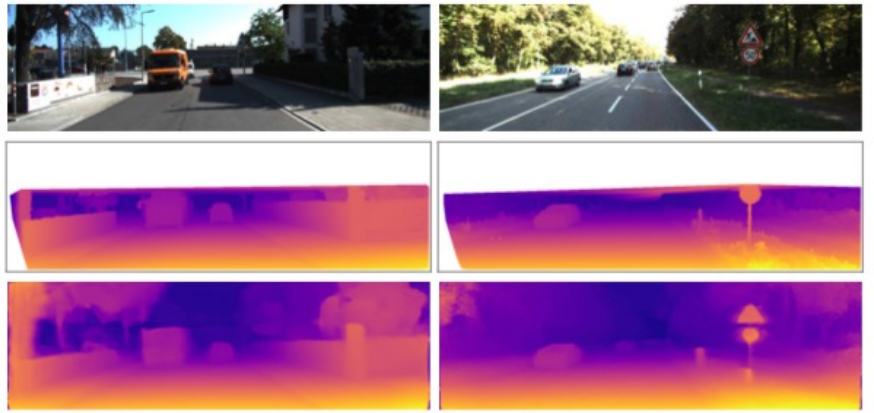


Figure 1. Our depth prediction results on KITTI 2015. Top to bottom: input image, ground truth disparities, and our result. Our method is able to estimate depth for thin structures such as street signs and poles.

Stereo Datasets

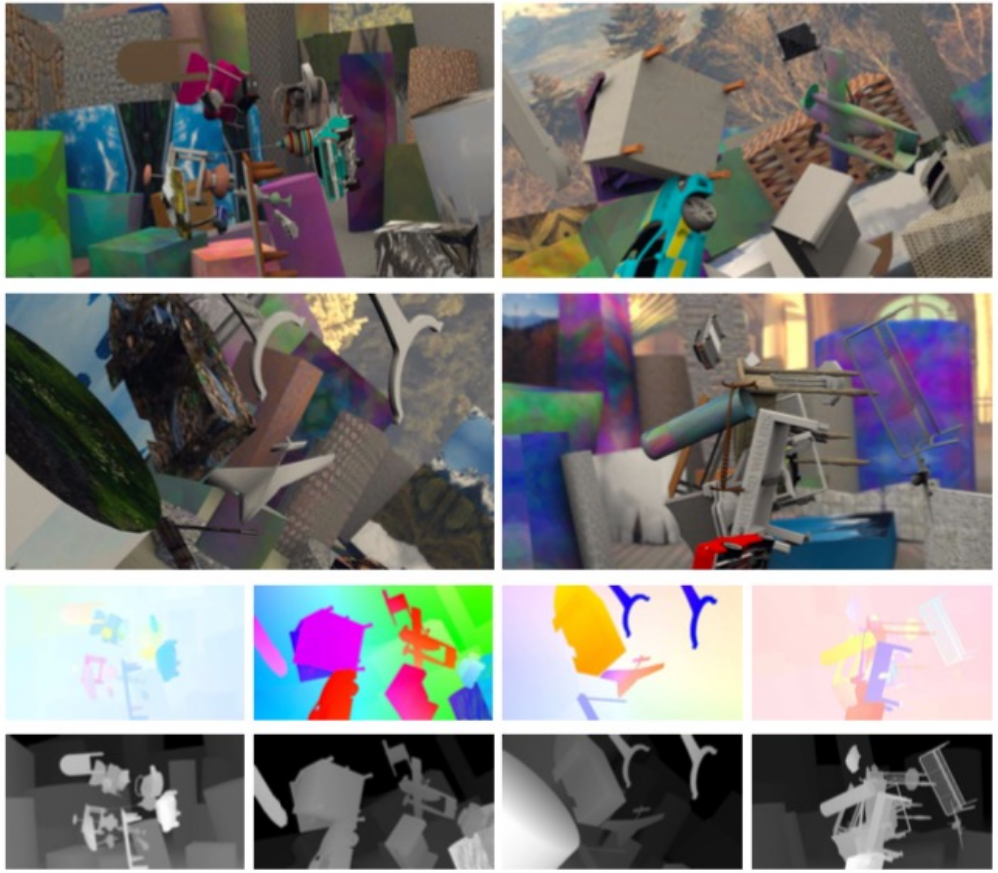
- *FlyingThings3D -> DispNet*

A Large Dataset to Train Convolutional Networks for Disparity, Optical Flow, and Scene Flow Estimation

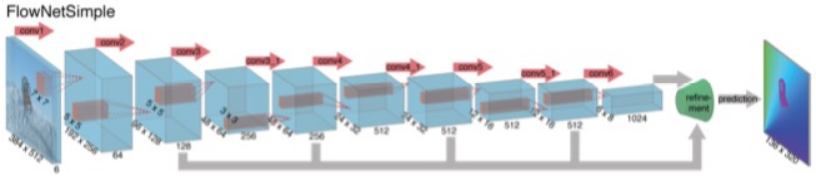
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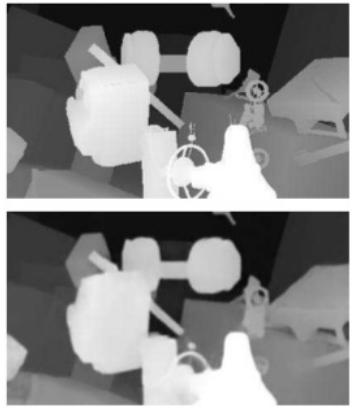
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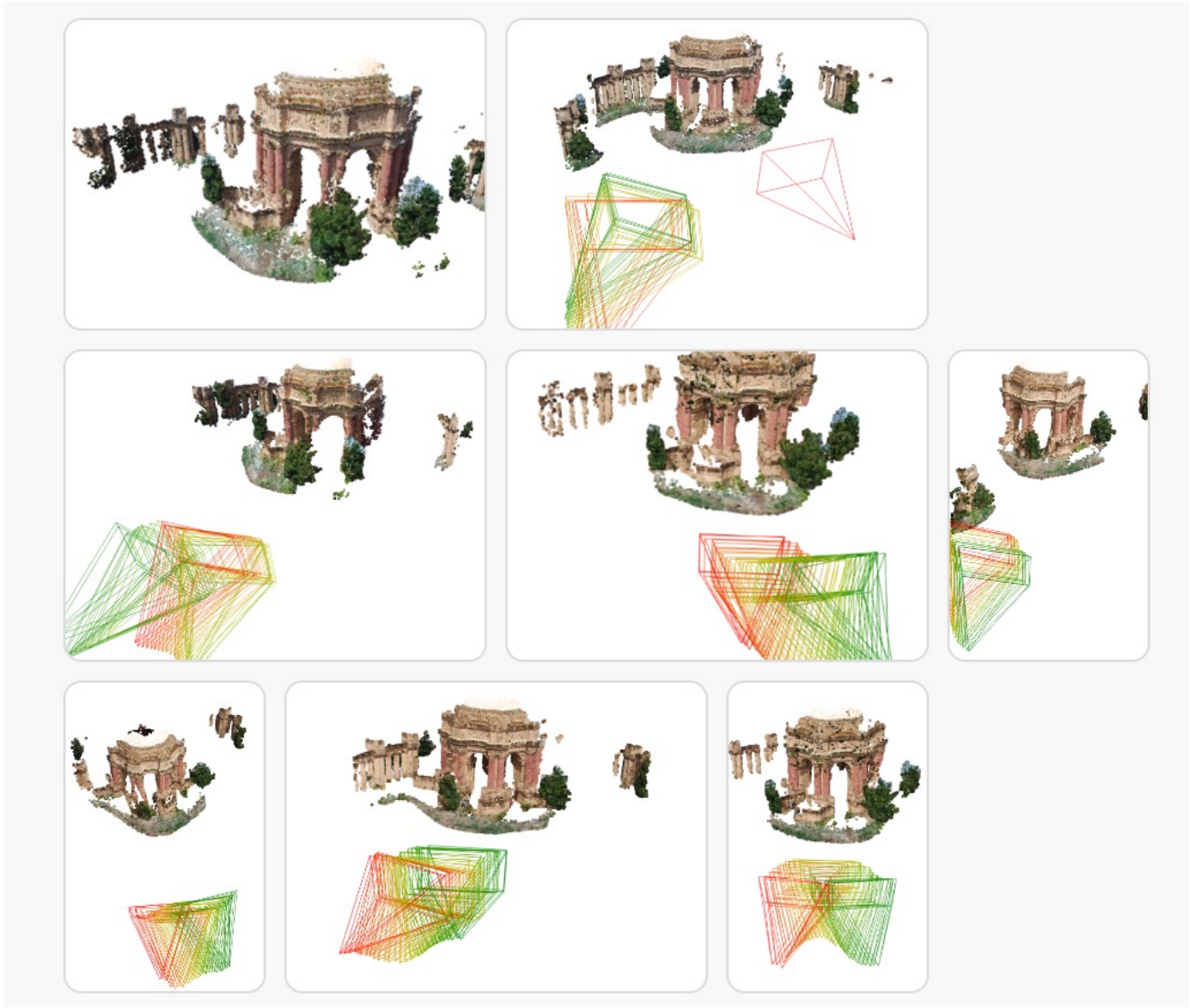




disp GT / prediction



3D Reconstruction VGGT: Best paper at CVPR 2025



Summary

5. Convolutional Neural Nets
6. Visualizing Learned Filters
7. Pooling and FC Layers
8. Applications in Vision
9. Object Classification
10. Applications in Robotics

